LECTURE NOTES
OF

STATISTICAL MECHANICS

Collection of the lectures notes of professor Orlandini.

Created By

Alice Pagano

Contributors:
Carlo Sgorlon Gaiatto
Noemi Manara
Jacopo Carotenuto

The University of Padua

Academic year 2023-2024

Source available: https://github.com/AlicePagano/Lecture-Notes-of-Statistical-Mechanics

Compiled: Tuesday 17th March, 2026

Contents

Introduction **v**

1 Recall of Thermodynamics **1**
1.1 A short recap of thermodynamics definitions . . . 1
1.2 Equilibrium states . . . 1
1.3 Equations of states . . . 3
1.4 Legendre transform and thermodynamic potentials . . . 4
1.5 Maxwell relations . . . 7
1.6 Response functions . . . 9
1.6.1 Response functions and thermodynamic stability . . . 10

2 Equilibrium phases and thermodynamics of phase transitions **13**
2.1 Equilibrium phases as minima of Gibbs free energy . . . 13
2.2 First order phase transition and phase coexistence . . . 13
2.2.1 Critical points . . . 18
2.2.2 Ferromagnetic system . . . 18
2.3 Second order phase transition . . . 21
2.3.1 Helmholtz free-energy . . . 22
2.4 Thermodynamic of phase coexistence . . . 22
2.4.1 Lever Rule . . . 22
2.4.2 Phase coexistence (one component system) . . . 23
2.4.3 Clausius-Clapeyron equation . . . 24
2.4.4 Application of C-C equation to the liquid-gas coexistence line . . . 25
2.5 Order parameter of a phase transition . . . 26
2.6 Classification of the phase transitions . . . 28
2.6.1 Thermodynamic classification . . . 28
2.6.2 Eherenfest classification . . . 28
2.6.3 Modern classification . . . 29
2.7 Critical exponents . . . 30
2.7.1 Divergence of the response functions at the critical point . . . 30
2.7.2 Critical exponents definition . . . 30
2.7.3 Law of the corresponding states (Experimental estimates of order parameters) . . . 32
2.7.4 Thermodynamic inequalities between critical exponents . . . 32

3 Recall of statistical mechanics and theory of ensembles **35**
3.1 Statistical ensembles . . . 35
3.1.1 Microcanonical ensemble . . . 35
3.2 The canonical ensemble . . . 36
3.2.1 Energy fluctuations in the canonical ensemble . . . 38
3.3 Isothermal and isobaric ensemble . . . 39
3.3.1 Saddle point approximation . . . 41
3.4 Gran canonical ensemble . . . 41

4 Statistical mechanics and phase transitions **43**
4.1 Statistical mechanics of phase transitions . . . 43
4.1.1 Magnetic system (canonical) . . . 44
4.1.2 Fluid system (gran canonical) . . . 44

4.1.3 Partition Function as Generating Function . . . 45
4.1.4 Thermodynamic limit with additional constraints . . . 45
4.1.5 Statistical mechanics and phase transitions . . . 45
4.2 Critical point and correlations of fluctuations . . . 46
4.2.1 Critical opalescence: experimental evidence of correlation length divergence . . . 48
4.3 Finite size effects and phase transitions . . . 48
4.4 Numerical simulations and phase transitions . . . 49

5 Role of the models in statistical mechanics **51**
5.1 Role of the models . . . 51
5.2 The Ising model . . . 51
5.2.1 d-dimensional Ising model . . . 51
5.2.2 Thermodynamic Limit . . . 53
5.2.3 Order Parameter of the Ising Model . . . 54
5.2.4 Ising model and $\mathbb{Z}^2$ symmetry. . . . 54

Disclaimer

Check all the formulas against the professor-supplied material. Care has been taken to be as accurate as possible, but errors are possible.

Introduction

The goal of statistical mechanics is to predict the macroscopic properties of bodies, most especially their thermodynamic properties, on the basis of their microscopic structure. The macroscopic properties of greatest interest to statistical mechanics are those relating to thermodynamic equilibrium. As a consequence, the concept of thermodynamic equilibrium occupies a central position in the field.

The microscopic structure of systems examined by statistical mechanics can be described by means of mechanical models: for example, gases can be represented as systems of particles that interact by means of a phenomenologically determined potential. Other examples of mechanical models are those that represent polymers as a chain of interconnected particles, or the classical model of crystalline systems, in which particles are arranged in space according to a regular pattern, and oscillate around the minimum of the potential energy due to their mutual interaction. The models to be examined can be, and recently increasingly are, more abstract, however, and exhibit only a faint resemblance to the basic mechanical description (more specifically, to the quantum nature of matter). The explanation of the success of such abstract models is itself the topic of one of the more interesting chapters of statistical mechanics: the theory of universality and its foundation in the renormalization group.

The models of systems dealt with by statistical mechanics have some common characteristics. We are in any case dealing with systems with a large number of degrees of freedom: the reason lies in the corpuscular (atomic) nature of matter. The degrees of freedom that one considers should have more or less comparable effects on the global behaviour of the system. This state of affairs excludes the application of the methods of statistical mechanics to cases in which a restricted number of degrees of freedom "dominates" the others—for example, in celestial mechanics, although the number of degrees of freedom of the planetary system is immense, an approximation in which each planet is considered as a particle is a good start. In this case, we can state that the translational degrees of freedom (three per planet)—possibly with the addition of the rotational degrees of freedom, also a finite number—dominate all others. These considerations also make attempts to apply statistical concepts to the human sciences problematic because, for instance, it is clear that, even if the behaviour of a nation's political system includes a very high number of degrees of freedom, it is possible to identify some degrees of freedom that are disproportionately important compared to the rest.

On the other hand, statistical methods can also be applied to systems that are not strictly speaking mechanical—for example, neural networks (understood as models of the brain's components), urban thoroughfares (traffic models), or problems of a geometric nature (percolation).

The simplest statistical mechanical model is that of a large number of identical particles, free of mutual interaction, inside a container with impenetrable and perfectly elastic walls. This is the model of the ideal gas, which describes the behaviour of real gases quite well at low densities, and more specifically allows one to derive the well-known equation of state.

The introduction of pair interactions between the particles of the ideal gas allows us to obtain the standard model for simple fluids. Generally speaking, this model cannot be resolved exactly and is studied by means of perturbation or numerical techniques. It allows one to describe the behaviour of real gases (especially noble gases), and the liquid–vapor transition (boiling and condensation).

The preceding models are of a classical (non-quantum) nature and can be applied only when the temperatures are not too low. The quantum effects that follow from the inability to distinguish particles are very important for phenomenology, and they can be dealt with at the introductory level if one omits interactions between particles.

In many of the statistical models we will describe, however, the system's fundamental elements will not be "particles," and the fundamental degrees of freedom will not be mechanical (position and velocity or impulse). If we want to understand the origin of ferromagnetism, for example, we should isolate only those degrees of freedom that are relevant to the phenomenon being examined (the orientation of the electrons' magnetic moment) from all those that are otherwise pertinent to the material in question.

The simplest case is that in which there are only two values—in this fashion, we obtain a simple model of ferromagnetism, known as the Ising model, which is by far the most studied model in statistical mechanics. The ferromagnetic solid is therefore represented as a regular lattice in space, each point of which is associated with a degree of freedom, called spin, which can assume the values +1 and -1. This model allows one to describe the paramagnet-ferromagnet transition, as well as other similar transitions.

In this course, classical statistical mechanics of system at equilibrium is treated.

Chapter 1

Recall of Thermodynamics

1.1 A short recap of thermodynamics definitions

The systems we are considering are

1. In equilibrium with an external bath at fixed temperature T.
2. Made by a (large) number N of degrees of freedom. For instance, we remind that $1\,\text{mol} \approx N_A \sim 10^{23}$ elementary units.

Thermodynamic is a macroscopic theory of matter at equilibrium. It starts either from experimental observations or from *assiomatic assumptions* and establishes rigorous relations between macroscopic variables (**observables**) to describe systems at equilibrium. One of the first important concepts is the one of **extensive variables**. For instance, the *extensive variables* that characterize the system at equilibrium are the internal energy U, volume V, number of particles N and magnetization $\vec{\mathbf{M}}$ that "scale with the system". In general, the extensive variable are *additive*.

In thermodynamics, it is important the concept of **walls and thermodynamic constrains** that are necessary for a complete definition of a thermodynamic system. With their presence or absence it is possible to control and redistribute the thermodynamic variables for changing the system. The typical walls are:

- *Adiabatic walls*: no heat flux. If it is removed we obtain a *diathermic walls*.
- *Rigid walls*: no mechanical work. If it is removed we obtain a *flexible or mobile walls*.
- *Impermeable walls*: no flux of particles (the number of particles remain constant). If it is removed we obtain a *permeable walls*.

1.2 Equilibrium states

Consider a system in an equilibrium state, if the system changes our aim is to study the next equilibrium state of the system. Therefore, we move from a system in equilibrium to another. The fundamental problem of thermodynamics is how to characterize the new system.

Now, we define the concept of **equilibrium states**. Consider macroscopic states that are fully described by extensive variables such as the internal energy U, the volume V, the number of particles N, the magnetization $\vec{\mathbf{M}}$, etc .... If these variables are time independent, the system is in a *steady state*. Moreover, if there are no macroscopic currents, the system is at *equilibrium*. Therefore, we describe a system by characterizing all the extensive variables at equilibrium.

Suppose that the system changes slow in time, it goes from an equilibrium state to another one and the transformation is so slow that in each Δt the system is at equilibrium. Hence, considering a sequence of equilibrium states, the **quasi-static transformation** are described by the 1st *Law of Thermodynamic*:

$$\mathrm{d}U = \delta Q - \delta W \tag{1.1}$$

The variation of the internal energy of the systems depends by two factors: δw that is the work done by the system during a *quasi-static process* (infinitive slow), and δQ that is the heat absorbed by the system

during the process. Remember that we write $\mathrm{d}U$ because it is a differential quantity, while the other quantities with the δ are only small quantities. Therefore, $\mathrm{d}U$ is a function of state, the other are not.

Remark. The convention is $\delta Q > 0$ if the heat is absorbed by the system, and $\delta W > 0$ if the work is done by the system.

For example, considering a simple fluid with a given pressure, if we change the volume, the work done by the systems is $\delta W = P\,\mathrm{d}V$. For a magnetized system, we have $\delta W = -\vec{\mathbf{H}} \cdot \mathrm{d}\vec{\mathbf{M}}$.

In conclusion, starting from an equilibrium state and removing some constraints (i.e. wall properties), we want to find the new equilibrium state compatible with the new constrains.

Suppose a system with adiabatic rigid impermeable constraints. The system on the left is characterized by V_1, N_1, U_1, the one on the right by V_2, N_2, U_2. There are many ways for solving this problem. We use the most general way, that is by using the *maximum entropy principle*. If there exists a function S of the extensive variables of the system that is defined for all equilibrium states, we call it **entropy** and the 1st *fundamental relation* is

$$S = S(U, V, N) \tag{1.2}$$

The new values taken by the extensive parameters when a constraint has been removed are the ones that *maximize* S. It means $\mathrm{d}S = 0$ and $\mathrm{d}^2S < 0$, given the remaining constraint.

The properties of S are:

1. S is an *additive function* with respect to the subsystems in which the system is partitioned:

$$S = \sum_{\alpha} S^{(\alpha)} \tag{1.3}$$

2. S is *differentiable* and *monotonically increasing* with respect to the internal energy U. It means that $\left(\frac{\partial S}{\partial U}\right)_{V,N} > 0$.

3. For each subsystem (α) we have:

$$S^{(\alpha)} = S^{(\alpha)}(U^{(\alpha)}, V^{(\alpha)}, N^{(\alpha)}) \tag{1.4}$$

   This fundamental relation holds for each subsystem.

4. S is an *homogeneous function* of 1st order with respect to the extensive parameters, namely:

$$S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N), \quad \forall \lambda > 0 \tag{1.5}$$

   It means that S is an extensive quantity.

Remark. Since S is monotonically increasing in U, the following inequality holds:

$$\left(\frac{\partial S}{\partial U}\right)_{V,N} > 0$$

Therefore, we have $\left(\frac{\partial S}{\partial U}\right)_{V,N} \neq 0$ and it can be inverted locally.

Afterwards, $S = S(U, V, N)$ inverted in U gives the 2st *fundamental relation*

$$U = U(S, V, N) \tag{1.6}$$

It means that, we can look or S or U and, when this quantities are known, all the informations about the system can be obtained.

By taking the differential of the fundamental relation

$$U = U(S, V, N_1, \ldots, N_r)$$

one gets

$$\mathrm{d}U = \underbrace{\left(\frac{\partial U}{\partial S}\right)_{V,N_j}}_{\substack{T \\ \text{absolute} \\ \text{temperature}}} \mathrm{d}S + \underbrace{\left(\frac{\partial U}{\partial V}\right)_{S,N_j}}_{\substack{-P \\ \text{pressure}}} \mathrm{d}V + \sum_{j=1}^{r} \underbrace{\left(\frac{\partial U}{\partial N_j}\right)_{S,V}}_{\substack{\mu_j \\ \text{electrochemical} \\ \text{potential}}} \mathrm{d}N_j \tag{1.7}$$

1.3 Equations of states

Now, we define another set of variables that are called **intensive variables**. The term *intensive* means that it is independent of the size of the system, namely that the value of the variable relative to a subsystem is equal to that of the whole system. The intensive variables are themselves functions of S, V, N, and examples of intensive variables are the pressure, P, and the temperature of the system, T.

The **state equations** are defined as:

$$T = T(S, V, N_1, \dots, N_r) \tag{1.8a}$$
$$P = P(S, V, N_1, \dots, N_r) \tag{1.8b}$$
$$\mu_j = \mu_j(S, V, N_1, \dots, N_r) \tag{1.8c}$$

Remark. If all the state equations are known, the fundamental relation is determined a part from a constant. It means that the coefficients of the differential (1.7) are known.

Example 1

Let us see some examples of equations of state:

- For an *ideal gas*:
$$PV = NK_bT \tag{1.9}$$
- *Van-Der Walls* equation of the state:
$$\left(P + \frac{\alpha N^2}{V^2}\right)(V - Nb) = Nk_bT \tag{1.10}$$
- For magnetic systems, another equation of state is the *Curie Law*:
$$M = \frac{CH}{T} \tag{1.11}$$

  Remark. We compute $\left(\frac{\partial U}{\partial M}\right)_{S,N} = H$.

The equations of state are homogeneous functions of *zero* degree. For example, considering the temperature T:

$$T(\lambda S, \lambda V, \lambda N) \overset{0}{=} T(S, V, N)$$

It means that at equilibrium the temperature of a subsystem is equal to the one of the whole system. Similarly,

$$P(\lambda S, \lambda V, \lambda N) = P(S, V, N)$$

Now, we keep the S parameter separates from the other that are substituted by *generalized displacements*, as $(V, N_1, \dots, N_r) \to X_j$. The fundamental relation becomes

$$U = U(S, X_1, \dots, X_{r+1}) \tag{1.12}$$

and we define:

$$\left(\frac{\partial U}{\partial S}\right) \equiv T \tag{1.13a}$$
$$\left(\frac{\partial U}{\partial X_j}\right) \equiv P_j \tag{1.13b}$$

The differential is written as the following:

$$\mathrm{d}U = T\,\mathrm{d}S + \sum_{j=1}^{r+1} P_j\,\mathrm{d}X_j \tag{1.14}$$

where $X_1 = V$ is the volume and $P_1 = -P$ is the pressure.

From the equilibrium condition,

$$\mathrm{d}U = 0$$

one can get a relation between intensive variables in differential form as the **Gibbs-Duhem relation**:

$$S\,\mathrm{d}T + \sum_{j=1}^{r+1} X_j\,\mathrm{d}P_j = 0 \tag{1.15}$$

For a one-component simple fluid system, the equation (1.15) simplifies into

$$S\,\mathrm{d}T - V\,\mathrm{d}P + N\,\mathrm{d}\mu = 0$$

and dividing by the number of moles N

$$\mathrm{d}\mu = -s\,\mathrm{d}T + v\,\mathrm{d}P \tag{1.16}$$

that is the Gibbs-Duhem relation in a *molar form*.
For a magnetic system, we have

$$\mathrm{d}U = T\,\mathrm{d}S + \vec{\mathbf{H}}\cdot\mathrm{d}\vec{\mathbf{M}} + \mu\,\mathrm{d}N \tag{1.17}$$

Remark. Note that $\mu = \mu(T, P)$ is a relation between intensive variables.

To summarize, the fundamental relations are $S = S(U, V, N_1, \dots, N_r)$ or $S = S(U, \vec{\mathbf{M}}, N_1, \dots, N_r)$ for magnetic systems. In the energy representation we have $U = U(S, V, N_1, \dots, N_r)$ or $U = U(S, \vec{\mathbf{M}}, N_1, \dots, N_r)$.

1.4 Legendre transform and thermodynamic potentials

In many situations, it is convenient to change some extensive variables with their conjugate intensive ones that became independent and free to vary. We have new thermodynamic potentials. It works as following; suppose we have a function as

$$Y = Y(X_0, X_1, \dots, X_k, \dots, X_{r+1}) \tag{1.18}$$

such that Y is *strictly convex* in say, X_k ($\frac{\partial^2 Y}{\partial X_k^2} > 0$) and smooth[1]. The idea is to find a transformation such that

$$Y = Y(X_0, X_1, \dots, P_k, \dots, X_{r+1}) \tag{1.19}$$

where

$$X_k \to P_k \equiv \frac{\partial Y}{\partial X_k} \tag{1.20}$$

i.e. P_k substitutes X_k as a new independent variable. In mathematics this is called *Legendre transform*.

The **thermodynamic potentials** are extremely useful tools, whose name derives from an analogy with mechanical potential energy: as we will later see, in certain circumstances the work obtainable from a macroscopic system is related to the change of an appropriately defined function, the thermodynamic potential. They are useful because they allow one to define quantities which are experimentally more easy to control and to rewrite the fundamental thermodynamic relations in terms of them.

Mathematically, *all the thermodynamic potentials are the result of a Legendre transformation of the internal energy*, namely they are a rewriting of the internal energy so that a variable has been substituted with another.

Example 2: How to calculate thermodynamic potentials

Suppose we want to replace the entropy S with its conjugate derivative

$$T = \frac{\partial U}{\partial S}$$

One starts form the fundamental relation

$$U = U(S, V, N_1, \dots)$$

[1] A smooth function is a function that has continuous derivatives up to some desired order over some domain.

and transforms U such that S is replaced by T as a new independent variable. Let us consider the transformation

$$A \equiv U - S\frac{\partial U}{\partial S} = U - TS$$

By differentiating A we get

$$\mathrm{d}A = \mathrm{d}U - T\,\mathrm{d}S - S\,\mathrm{d}T$$

On the other hand

$$\mathrm{d}U = T\,\mathrm{d}S + \sum_j P_j\,\mathrm{d}X_j$$

It implies that

$$\mathrm{d}A = -S\,\mathrm{d}T + \sum_j P_j\,\mathrm{d}X_j$$

For such a system we have $A = A(T, V, N_1, \ldots, N_r)$. It is a function of T instead of S, as wanted. Similarly for a magnetic system $A = A(T, \vec{\mathbf{M}}, N_1, \ldots, N_r)$.

Helmholtz free energy

The **Helmholtz free energy** is defined as:

$$A \equiv U - TS \tag{1.21}$$

In terms of heat and mechanical work, since $\mathrm{d}U = \delta Q - \delta W$:

$$\mathrm{d}A = \mathrm{d}U - \mathrm{d}(TS) = \delta Q - T\,\mathrm{d}S - S\,\mathrm{d}T - \delta W$$

Hence,

$$\delta W = (\delta Q - T\,\mathrm{d}S) - S\,\mathrm{d}T - \mathrm{d}A \tag{1.22}$$

On the other hand, for a *reversible transformation* we have

$$\delta Q = T\,\mathrm{d}S$$

which implies

$$\delta W = -S\,\mathrm{d}T - \mathrm{d}A \tag{1.23}$$

If the reversible transformation is also *isothermal*, $\mathrm{d}T = 0$ and we obtain $\mathrm{d}A = \mathrm{d}W$. It is reminiscent of a potential energy.

Remark. For an isothermal but *not reversible* (spontaneous) process we know the 2nd *Law of Thermodynamics*

$$\delta Q \leq T\,\mathrm{d}S$$

which implies

$$(\delta W)_{irr} = \delta Q - T\,\mathrm{d}S - \mathrm{d}A \leq -\,\mathrm{d}A\,. \tag{1.24}$$

Hence, if $\delta W = 0$ and $\mathrm{d}T = 0$, we have $\mathrm{d}A \leq 0$. Therefore, in a spontaneous (irreversible) process, the thermodynamic system, as a function of T, V, N etc, evolves towards a *minimum* of the Helmoltz free energy $A = A(T, V, N_1, \ldots, N_r)$.

In the case of a system with (P, V, T), we have:

$$\mathrm{d}A = -S\,\mathrm{d}T - P\,\mathrm{d}V + \sum_j \mu_j\,\mathrm{d}N_j \tag{1.25}$$

where

$$-S = \left(\frac{\partial A}{\partial T}\right)_{V,N_j} \tag{1.26a}$$

$$-P = \left(\frac{\partial A}{\partial V}\right)_{T,N_j} \tag{1.26b}$$

$$\mu_j = \left(\frac{\partial A}{\partial N_j}\right)_{T,V} \tag{1.26c}$$

For a magnetic system $(\vec{\mathbf{H}}, \vec{\mathbf{M}}, T)$:

$$\mathrm{d}A = -S\,\mathrm{d}T + \vec{\mathbf{H}} \cdot \mathrm{d}\vec{\mathbf{M}} + \sum_j \mu_j\,\mathrm{d}N_j \tag{1.27}$$

with

$$H_\alpha = \left(\frac{\partial A}{\partial M_\alpha}\right)_{T,\{N_j\}} \tag{1.28}$$

Heltalpy

The **Hentalpy** is the partial Legendre transform of U that replaces the volume V with the pressure P as independent variable.

Consider $U = U(S, V, N_1, \ldots, N_r)$ and $-P = \frac{\partial U}{\partial V}$, we define the hentalpy as

$$H = U + PV \tag{1.29}$$

Remark. Note that the plus sign in the definition of the hentalpy is just because the minus of the P.

We have:

$$\begin{aligned} \mathrm{d}H &= \mathrm{d}U + P\,\mathrm{d}V + V\,\mathrm{d}P \\ &= T\,\mathrm{d}S - \cancel{P\,\mathrm{d}V} + \sum_j \mu_j\,\mathrm{d}N_j + \cancel{P\,\mathrm{d}V} + V\,\mathrm{d}P \\ &= T\,\mathrm{d}S + V\,\mathrm{d}P + \sum_j \mu_j\,\mathrm{d}N_j \end{aligned} \tag{1.30}$$

Finally, we obtain the relation $H = H(S, P, N_1, \ldots, N_r)$.

Gibbs potential

The **Gibss potential** is obtained by performing the Legendre transform of U to replace S and V with T and P.

Consider again $U = U(S, V, N_1, \ldots, N_r)$ and $T = \frac{\partial U}{\partial S}, -P = \frac{\partial U}{\partial V}$, then we have:

$$G = U - TS + PV = A + PV \tag{1.31}$$

For a simple fluid system

$$\begin{aligned} \mathrm{d}G &= \mathrm{d}U - T\,\mathrm{d}S - S\,\mathrm{d}T + P\,\mathrm{d}V + V\,\mathrm{d}P \\ &= \cancel{T\,\mathrm{d}S} - \cancel{P\,\mathrm{d}V} + \sum_j \mu_j\,\mathrm{d}N_j - \cancel{T\,\mathrm{d}S} - S\,\mathrm{d}T + \cancel{P\,\mathrm{d}V} + V\,\mathrm{d}P \\ &= -S\,\mathrm{d}T + V\,\mathrm{d}P + \sum_j \mu_j\,\mathrm{d}N_j \end{aligned} \tag{1.32}$$

Hence, $G = G(T, P, N_1, \ldots, N_r)$.

For a magnetic system, the Gibbs potential is defined as

$$G = A - \vec{\mathbf{M}} \cdot \vec{\mathbf{H}} \tag{1.33}$$

and

$$\begin{aligned} \mathrm{d}G &= \mathrm{d}A - \mathrm{d}\vec{\mathbf{M}} \cdot \vec{\mathbf{H}} = \mathrm{d}(U - TS) - \mathrm{d}(-\vec{\mathbf{M}} \cdot \vec{\mathbf{H}}) \\ &= \mathrm{d}U - T\,\mathrm{d}S - S\,\mathrm{d}T - \mathrm{d}\vec{\mathbf{M}} \cdot \vec{\mathbf{H}} - \vec{\mathbf{M}} \cdot \mathrm{d}\vec{\mathbf{H}} \\ &= \cancel{T\,\mathrm{d}S} + \cancel{\vec{\mathbf{H}} \cdot \mathrm{d}\vec{\mathbf{M}}} - \cancel{T\,\mathrm{d}S} - S\,\mathrm{d}T - \cancel{\mathrm{d}\vec{\mathbf{M}} \cdot \vec{\mathbf{H}}} - \vec{\mathbf{M}} \cdot \mathrm{d}\vec{\mathbf{H}} \\ &= -S\,\mathrm{d}T - \vec{\mathbf{M}} \cdot \mathrm{d}\vec{\mathbf{H}} \end{aligned} \tag{1.34}$$

and finally $G = G(T, \vec{\mathbf{H}})$ and also

$$S = -\left(\frac{\partial G}{\partial T}\right)_{\vec{\mathbf{H}}} \tag{1.35a}$$

$$\vec{\mathbf{M}} = -\left(\frac{\partial G}{\partial \vec{\mathbf{H}}}\right)_{T} \tag{1.35b}$$

Grand canonical potential

The **grand canonical potential** is obtained by performing the Legendre transform of U to replace S and N with T and μ. The corresponding Legendre transform is

$$\Omega = U - TS - \sum_{i=1}^{r} \mu_i N_i = A - \sum_{i=1}^{r} \mu_i N_i \tag{1.36}$$

Differentiating this relation we obtain:

$$\begin{aligned} \mathrm{d}\Omega &= \mathrm{d}U - S\,\mathrm{d}T - T\,\mathrm{d}S - \sum_{ij} \mathrm{d}\mu_j\, N_j - \sum_{i=1}^{r} \mu_i\,\mathrm{d}N_i \\ &= (\delta Q - T\,\mathrm{d}S) - \delta W - S\,\mathrm{d}T - \sum_{j=1}^{r} \mathrm{d}\mu_j\, N_j - \sum_{j=1}^{r} \mu_j\,\mathrm{d}N_j \end{aligned} \tag{1.37}$$

Hence, $\Omega = \Omega(T, P, \{\mu_j\})$.

1.5 Maxwell relations

Internal energy U and entropy S are homogeneus function of the first order. A consequence of this fact is the relation called **Euler equation**:

$$U = TS - PV + \sum_{j} \mu_j N_j \tag{1.38}$$

Example 3: How to derive the Euler equation

Using the additive property of the internal energy U, we can derive a useful thermodynamic relation, the Euler equation.

$$U(\lambda S, \lambda V, \lambda N_1, \dots, \lambda N_m) = \lambda U(S, V, N_1, \dots N_m)$$

Let us differentiate this "extensivity condition" with respect to λ:

$$\frac{\partial U(\lambda S, \dots)}{\partial(\lambda S)} S + \frac{\partial U(\lambda S, \dots)}{\partial(\lambda V)} V + \sum_{i=1}^{m} \frac{\partial U(\lambda S, \dots)}{\partial(\lambda N_i)} N_i = U(S, V, N_1, \dots, N_m)$$

Setting $\lambda = 1$ in the above equation, we obtain:

$$\frac{\partial U}{\partial S} S + \frac{\partial U}{\partial V} V + \frac{\partial U}{\partial N_1} N_1 + \dots + \frac{\partial U}{\partial N_m} N_m = U$$

Using the definition of the intensive parameters, we arrive at the Euler equation:

$$U = TS - PV + \sum_{i=1}^{m} \mu_i N_i$$

Instead, the **Maxwell relations** are relations between the mixed derivatives of the thermodynamic potentials. They can be obtained from the expressions of $\mathrm{d}U$, $\mathrm{d}H$, $\mathrm{d}A$, $\mathrm{d}G$ and $\mathrm{d}\Omega$ and from the Schwarz theorem on mixed partial derivatives.

Due to Schwarz theorem, if a thermodynamic potential depends on $t + 1$ variables there will be $\frac{t(t+1)}{2}$ independent mixed derivatives.

Example 4: Internal energy $U = U(S, V, N)$

$$\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V + \mu\,\mathrm{d}N \tag{1.39}$$

where

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N} \qquad -P = \left(\frac{\partial U}{\partial V}\right)_{S,N}$$

It implies that

$$\frac{\partial^2 U}{\partial V \partial S} = \left(\frac{\partial T}{\partial V}\right)_{S,N} \underset{\substack{\text{from}\\ \text{Schwarz inequality}}}{=} \frac{\partial^2 U}{\partial S \partial V} = -\left(\frac{\partial P}{\partial S}\right)_{V,N}$$

therefore, we have the *1° Maxwell relation* :

$$\left(\frac{\partial T}{\partial V}\right)_{S,N} = -\left(\frac{\partial P}{\partial S}\right)_{V,N}$$

All the *3 Maxweel relations* obtained by the differential (1.39) with $t = 2$, for which we have $t+1 = 3$ and $\frac{t(t+1)}{2} = 3$ ($[S, V, N]$), are

$$(S, V): \qquad \left(\frac{\partial T}{\partial V}\right)_{S,N} = -\left(\frac{\partial P}{\partial S}\right)_{V,N} \tag{1.40a}$$

$$(S, N): \qquad \left(\frac{\partial T}{\partial N}\right)_{V,S} = \left(\frac{\partial \mu}{\partial S}\right)_{V,N} \tag{1.40b}$$

$$(V, N): \qquad -\left(\frac{\partial P}{\partial N}\right)_{S,V} = \left(\frac{\partial \mu}{\partial V}\right)_{S,N} \tag{1.40c}$$

Example 5: Helmholz $A = A(T, V, N)$

$$\mathrm{d}A = -S\,\mathrm{d}T - P\,\mathrm{d}V + \mu\,\mathrm{d}N \tag{1.41}$$

In this case the *3 Maxweel relations* ($[T, V, N]$) are

$$(T, V): \qquad \left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial P}{\partial T}\right)_{V,N} \tag{1.42a}$$

$$(T, N): \qquad -\left(\frac{\partial S}{\partial N}\right)_{T,V} = \left(\frac{\partial \mu}{\partial T}\right)_{V,N} \tag{1.42b}$$

$$(V, N): \qquad -\left(\frac{\partial P}{\partial N}\right)_{V,T} = \left(\frac{\partial \mu}{\partial V}\right)_{T,N} \tag{1.42c}$$

Example 6: Gibbs $G = G(T, P, N)$

$$\mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}P + \mu\,\mathrm{d}N \tag{1.43}$$

In this case the *3 Maxweel relations* ($[T, P, N]$) are

$$(T, P): \qquad -\left(\frac{\partial S}{\partial P}\right)_{T,N} = \left(\frac{\partial V}{\partial T}\right)_{P,N} \tag{1.44a}$$

$$(T, N): \qquad -\left(\frac{\partial S}{\partial N}\right)_{T,P} = \left(\frac{\partial \mu}{\partial T}\right)_{P,N} \tag{1.44b}$$

$$(P, N): \qquad \left(\frac{\partial V}{\partial N}\right)_{P,T} = \left(\frac{\partial \mu}{\partial P}\right)_{T,N} \tag{1.44c}$$

1.6 Response functions

Response functions are quantities that express how a system reacts when some external parameters are changed.

In fact, aim of most experiments is to measure the response of a thermodynamic system with respect to controlled variations of thermodynamic variables. Any observation is just the perturbation of a system and looking for the response. A list of the commonly used response functions is the following:

- Thermal response functions (heat capacities):
  - *Heat capacity at constant pressure*

$$C_P \equiv \left(\frac{\delta Q}{\mathrm{d}T}\right)_{P,N} \underset{\delta Q = T\mathrm{d}S}{=} T\left(\frac{\partial S}{\partial T}\right)_{P,N} \underset{-S=\left(\frac{\partial G}{\partial T}\right)_{P,N}}{=} -T\left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} \tag{1.45}$$

  - *Heat capacity at constant volume*

$$C_V \equiv \left(\frac{\delta Q}{\mathrm{d}T}\right)_{V,N} \underset{\delta Q = T\mathrm{d}S}{=} T\left(\frac{\partial S}{\partial T}\right)_{V,N} \underset{-S=\left(\frac{\partial A}{\partial T}\right)_{V,N}}{=} -T\left(\frac{\partial^2 A}{\partial T^2}\right)_{V,N} \tag{1.46}$$

  Remark. Heat capacity C_P is an extensive property. The associated intensive property is the specific heat capacity c_P, which is obtained by dividing the heat capacity of an object by its mass. However, hereinafter these quantities will be used interchangeably.

- Mechanical response functions:
  - *Thermal expansion coefficient at constant pressure*

$$\alpha_P \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_{P,N} \tag{1.47}$$

  - *Adiabatic compressibility*

$$k_S \equiv -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{S,N} \underset{V=\left(\frac{\partial H}{\partial P}\right)_{S,N}}{=} -\frac{1}{V}\left(\frac{\partial^2 H}{\partial P^2}\right)_{S,N} \tag{1.48}$$

  - *Isothermal compressibility*

$$k_T \equiv -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{T,N} \underset{V=\left(\frac{\partial G}{\partial P}\right)_{T,N}}{=} -\frac{1}{V}\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} \tag{1.49}$$

- Magnetic response functions:
  - *Isothermal susceptibility (d=1)*

$$\chi_T \equiv \left(\frac{\partial M}{\partial H}\right)_T \underset{M=-\left.\frac{\partial G}{\partial H}\right|_T}{=} -\left(\frac{\partial^2 G}{\partial H^2}\right)_T \tag{1.50}$$

    More generally, if $\vec{\mathbf{M}}, \vec{\mathbf{H}}$ we have a second order tensor

$$\chi_{\alpha\beta} = \left(\frac{\partial M_\alpha}{\partial H_\beta}\right)_T, \quad M_\alpha = -\left.\frac{\partial G}{\partial H_\alpha}\right|_T \Rightarrow \chi_{\alpha\beta} = \left.\frac{\partial^2 G}{\partial H_\beta \partial H_\alpha}\right|_T \tag{1.51}$$

Remark. Note that the response functions, when used with the Maxwell relations, allow to express observables usually inaccessible to experiments with measurable quantities.

Let us illustrate a lemma useful for calculation:

Lemma 1

Let x, y, z be quantities that satisfy the relation $f(x, y, z) = 0$. If w is a function of two any variables chosen between x, y, z, then:

1. $\left(\frac{\partial x}{\partial y}\right)_w \left(\frac{\partial y}{\partial z}\right)_w = \left(\frac{\partial x}{\partial z}\right)_w$
2. $\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{\left(\frac{\partial y}{\partial x}\right)_z}$
3. $\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$ *(concatenation relation or triple product rule).*

Example 7

The Maxwell relation

$$\left(\frac{\partial S}{\partial P}\right)_{T,N} = -\left(\frac{\partial V}{\partial T}\right)_{P,N}$$

obtained from

$$\mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}P$$

((T, P) equation) used with the response function α_P, permits to write

$$\underbrace{\left(\frac{\partial S}{\partial P}\right)_{T,N}}_{\substack{\text{inaccessible}\\ \text{to experiments}}} = \underbrace{-V\alpha_P}_{\text{measurable}} \tag{1.52}$$

Example 8

Let us start with the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial P}{\partial T}\right)_{V,N}$$

obtained from ((T, V) equation)

$$\mathrm{d}A = -S\,\mathrm{d}T - P\,\mathrm{d}V$$

From some property of multi-variable differential calculus one has the *triple product rule*:

$$\left(\frac{\partial P}{\partial T}\right)_{V,N} \left(\frac{\partial V}{\partial P}\right)_{T,N} \left(\frac{\partial T}{\partial V}\right)_{P,N} = -1$$

Hence

$$\begin{aligned}\left(\frac{\partial P}{\partial T}\right)_{V,N} &= \frac{-1}{\left(\frac{\partial V}{\partial P}\right)_{T,N}\left(\frac{\partial T}{\partial V}\right)_{P,N}} = -\frac{\left(\frac{\partial V}{\partial T}\right)_{P,N}}{\left(\frac{\partial V}{\partial P}\right)_{T,N}} \\ &= \frac{-V\alpha_P}{-Vk_T} = \frac{\alpha_P}{k_T}\end{aligned} \tag{1.53}$$

1.6.1 Response functions and thermodynamic stability

Now, we analyze the concept of *thermal stability*. If one injects heat in a system either at constant volume or at constant pressure, its temperature will inevitably increase

$$\begin{cases} c_V \equiv \left(\frac{\delta Q}{\mathrm{d}T}\right)_V \geq 0 \\ c_P \equiv \left(\frac{\delta Q}{\mathrm{d}T}\right)_P \geq 0 \end{cases} \tag{1.54}$$

Remark. The thermal capacities are *non-negative functions*!

It is useful also the concept of *mechanical stability*. If one compress a system by keeping T constant, we would expect that it shrinks

$$k_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \geq 0 \tag{1.55}$$

Similar considerations for a magnetic system, gives

$$c_H \geq 0, \quad c_M \geq 0, \quad \chi_T \geq 0 \tag{1.56}$$

Remark. In diamangetic systems χ_M can also be negative.

Exercise 1

By using Maxwell relations show that

$$c_P - c_V = \frac{TV\alpha_P^2}{k_T} = \frac{1}{Vk_T}T\left(\frac{\partial V}{\partial T}\right)_P^2 \tag{1.57a}$$

$$c_H - c_M = \frac{T}{\chi_T}\left(\frac{\partial M}{\partial T}\right)_H^2 \tag{1.57b}$$

Solution. Let us start considering a system with a fixed number of particles (namely $\mathrm{d}N = 0$) and such that S is explicitly expressed in terms of T and V. Then:

$$\mathrm{d}S = \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V$$

Dividing by $\mathrm{d}T$ both sides keeping the pressure constant, and then multiplying by T:

$$T\left(\frac{\partial S}{\partial T}\right)_P - T\left(\frac{\partial S}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P$$

it implies

$$c_P - c_V = T\left(\frac{\partial S}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P$$

Now, using the Maxwell relation $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$ and using the triple product rule we obtain

$$\left(\frac{\partial P}{\partial T}\right)_V = -\left(\frac{\partial P}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P$$

we get:

$$c_P - c_V = -T\left(\frac{\partial P}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P^2 = -T\alpha_P^2 V^2\left(\frac{\partial P}{\partial V}\right)_T = \frac{TV}{k_T}\alpha_P^2$$

It can be shown similarly for magnetic systems.

A consequence is that, since the right hand terms are non negative, it follows that

$$\begin{cases} c_P \geq c_V \geq 0 \\ c_H \geq c_M \geq 0 \end{cases} \tag{1.58}$$

For resuming, we have seen the thermodynamics of a phase, where the equilibrium state can be described by the maximum of the entropy. If we have a given phase, we can look for the Gibbs function. If we have more phases, we want to change between these phases.

Chapter 2

Equilibrium phases and thermodynamics of phase transitions

2.1 Equilibrium phases as minima of Gibbs free energy

Experimentally, any element or compound can be found, depending on the thermodynamic conditions in which it is, in different phases. When we say that a system is in a particular phase we mean that its physical properties (like density or magnetization) are uniform.

Equilibrium states are given by *maxima* of the entropy and *minima* of internal energy, or by *minima* of thermodynamics potentials such as A and G. Let us consider for example the Gibbs potential per particle of a fluid system

$$\frac{G}{N} \equiv g = g(T, P) \tag{2.1}$$

that depends on two intensive variables T and P and is not anymore a function of N because we have divided for N. Let us define α as the phase of a one-component system (say $\alpha =$ gas or liquid). Therefore, the thermodynamic properties are described by surfaces of function $g_\alpha(T, P)$ and for all equilibrium phase we have a surface on the space (T,P,g). For each value of T and P the thermodynamically stable phase is the one for which $g_\alpha(T, P)$ is minimum.

2.2 First order phase transition and phase coexistence

Let us suppose for example that the system can be found in two phases α and β (for example liquid and solid). Consider the surface g_α and g_β, we are looking for the lower one.

For given values of T and P the stable phase will be that with the lowest value of g: for example, if we have $g_\alpha(T, P) < g_\beta(T, P)$ then the system will be in phase α. Therefore there will be regions in (T, P) space were the most stable phase will be α and others in which it will be β. If we now plot the values of g as a function of T and T in (g, P, T) space for every phase of the system, we can determine the regions where the two phases will be the stable ones, namely we can determine the phase diagram of the system, as illustrated in Figure 2.1.

The very interesting region of this space (and the one on which we will focus our attention in this section) is the line where the surfaces of the two phases intersect: along this the two phases coexist, and when the system crosses it we say that it undergoes a *phase transition*. The coexistence line is the projection on the *(T,P)* plane of the intersection between different surfaces, so the *coexistence condition* is:

$$g_\alpha(T, P) = g_\beta(T, P) \tag{2.2}$$

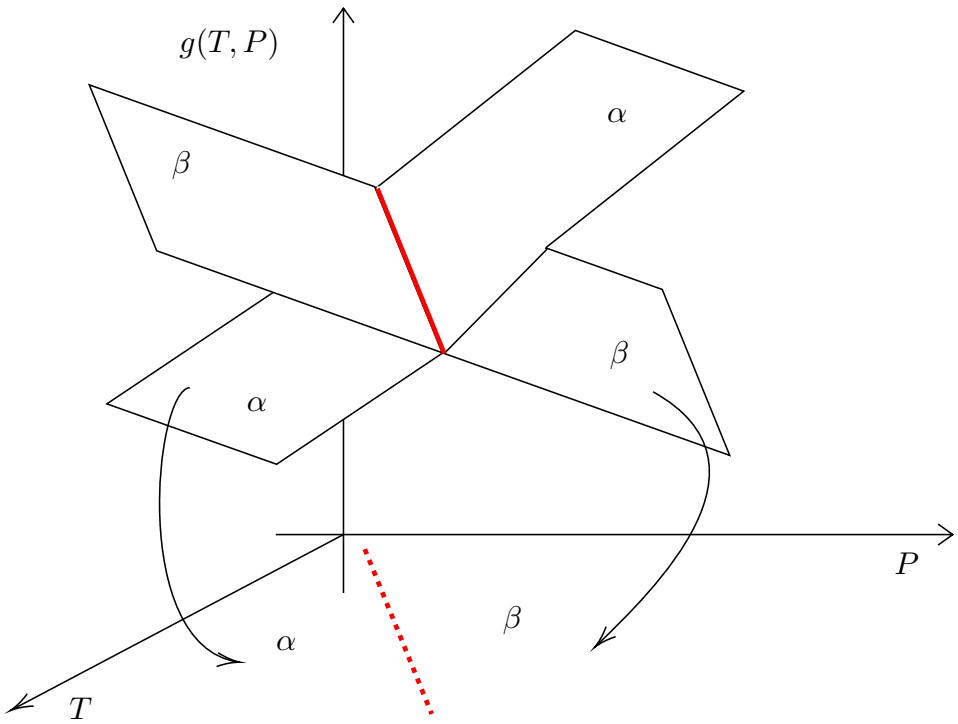


Figure 2.1: Phase diagram: stability of phases.

To fix the ideas, let us choose a given value of pressure $P = P^*$ and study the behavior of $g(T, P^*)$ as a function of T when we go from solid to gas, as illustrated in Figure 2.2.

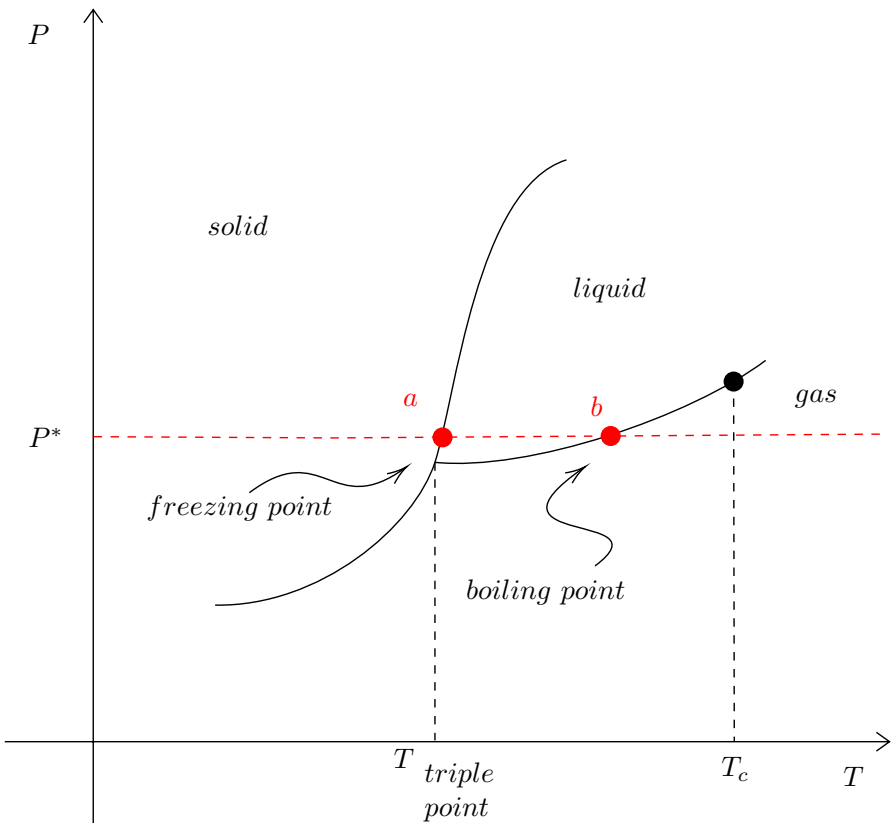


Figure 2.2: (T, P) projection.

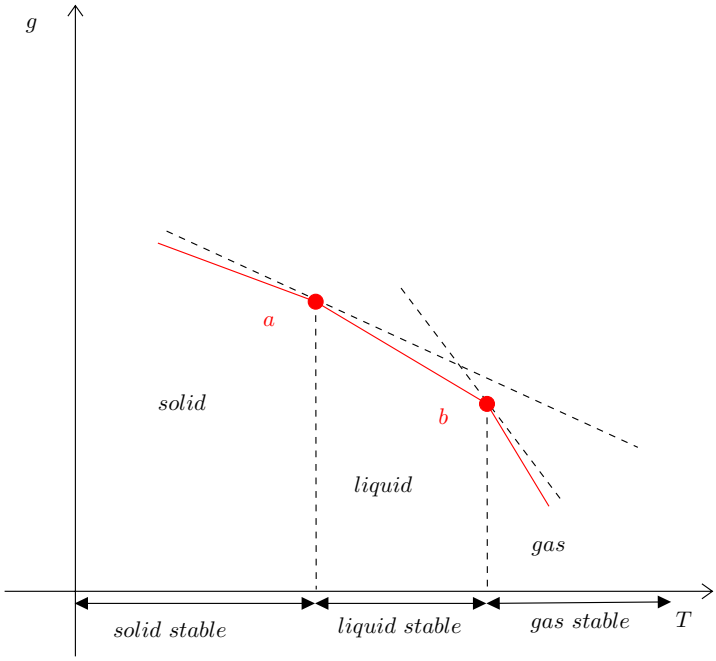


Figure 2.3: (g, T) projection at a fixed pression $P = P^*$. The red line is the coexistence line of the two phases α and β.

The existence of a critical point has a very intriguing consequence: since the liquid-gas coexistence line ends in a point, this means that a liquid can continuously be transformed in a gas (or viceversa), and in such a way that the coexistence of liquid and gaseous phases is never encountered.

At the coexistence line, $g_{\text{solid}}(T_a, P^*) = g_{\text{liq}}(T_a)$ and $g_{\text{liq}}(T_b) = g_{\text{gas}}(T_b, P^*)$, as shown in Figure 2.3.

Note also that:

- At the coexistence points a and b of the two phases, one has $g_\alpha(T) = g_\beta(T)$.

- $g(T)$ is a continuous function of T.

- Note that, $S = -\left(\frac{\partial G}{\partial T}\right)_V$ and $c_P = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_P > 0$. This implies that $g(T)$ is concave in T at fixed P.

How about its derivatives? Since P is fixed we can vary T and look for $s = -\left(\frac{\partial g}{\partial T}\right)_P$. As we cross different phases we have discontinuities, where $\Delta s T$ is called the *latent heat*. It is illustrated in Figure 2.4.

If there is a finite discontinuity in one, or more, of the *first* derivatives of the appropriate thermodynamic potential, the transition is called *first order transition*. In general, *a phase transition is signaled by a singularity in a thermodynamic potential.*

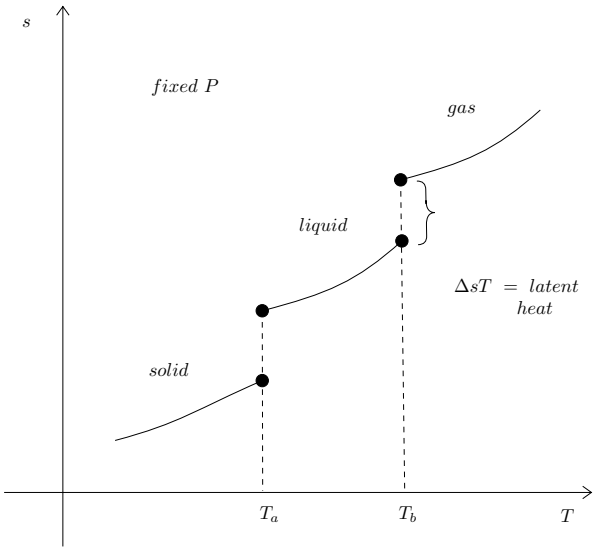


Figure 2.4: (s, T) projection.

We can also fix the temperature $T = T^*$ and look at the variation of P, as shown in Figure 2.5.

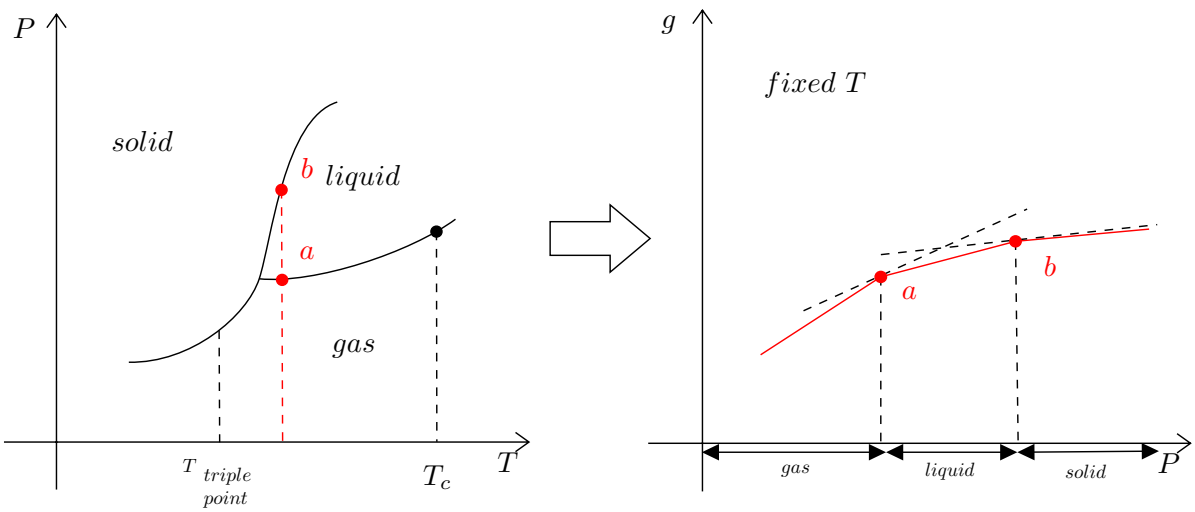


Figure 2.5: Left: (T, P) projection. Right: (g, P) projection at a fixed temperature $T = T^*$.

Note that, we have $v = \left(\frac{\partial g}{\partial P}\right)_T > 0$:

$$\left(\frac{\partial^2 g}{\partial P^2}\right) = \left(\frac{\partial v}{\partial P}\right)_T = -vk_T < 0 \tag{2.3}$$

so, also in this case we had a jump of the first order derivative of the thermodynamic potential g. It is illustrated in Figure 2.6.

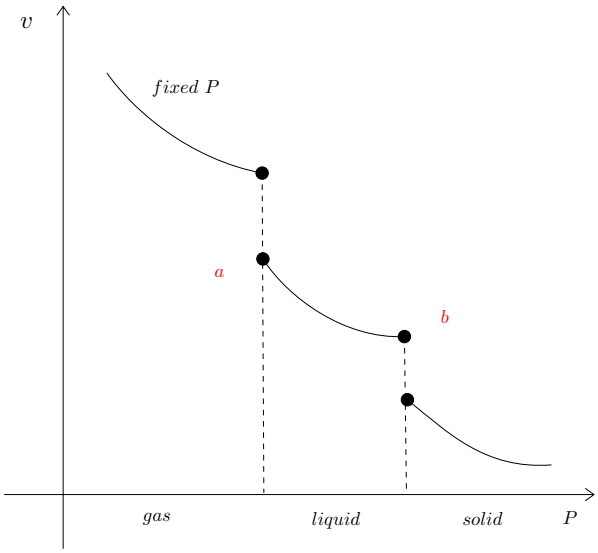


Figure 2.6: (v, P) projection.

2.2.1 Critical points

At the critical point (P_c, T_c) the system can pass from the liquid to the gas phase (and vice versa) in a continuous way

$$\Delta s = \Delta v = 0$$

Usually, critical points are end point of first order transition phases. Why there is no critical point between solid and liquid? The crossover between phases having the same symmetry define the Landau point. There is a break of symmetry, for instance we can think about the structure of the bravais lattice. Instead, from gas to liquid symmetries are not broken.

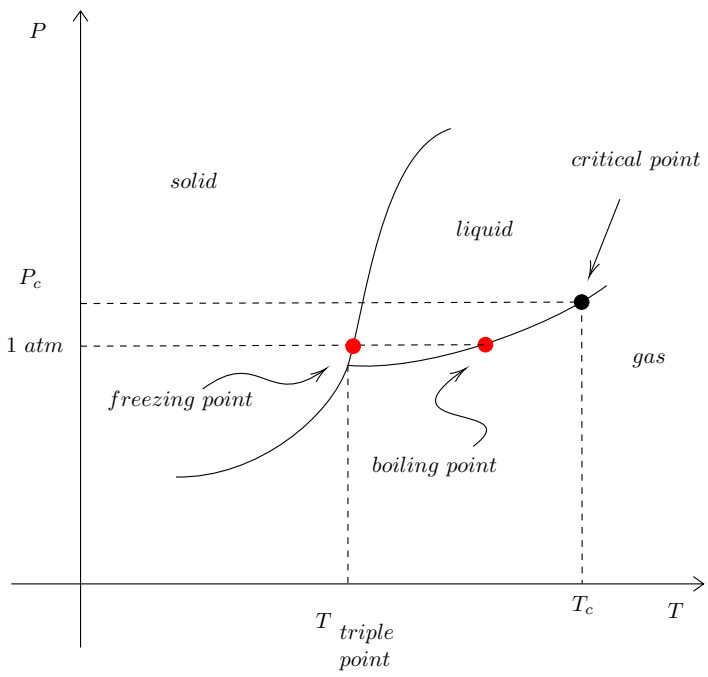


Figure 2.7: Phase diagram of a fluid. All the phase transition are first-order except at the critical point C. Beyond C it is possible to move continuously from liquid to a gas. The boundary between the solid and liquid phases is thought to be always first-oder and not to terminate in a critical point.

2.2.2 Ferromagnetic system

A similar behaviour can be encountered in magnetic systems. We can have a magnetization different from 0 even when the is no magnetic field. Supposing $P \leftrightarrow H, V \leftrightarrow M$, we have $(P, T) \leftrightarrow (H, T)$.

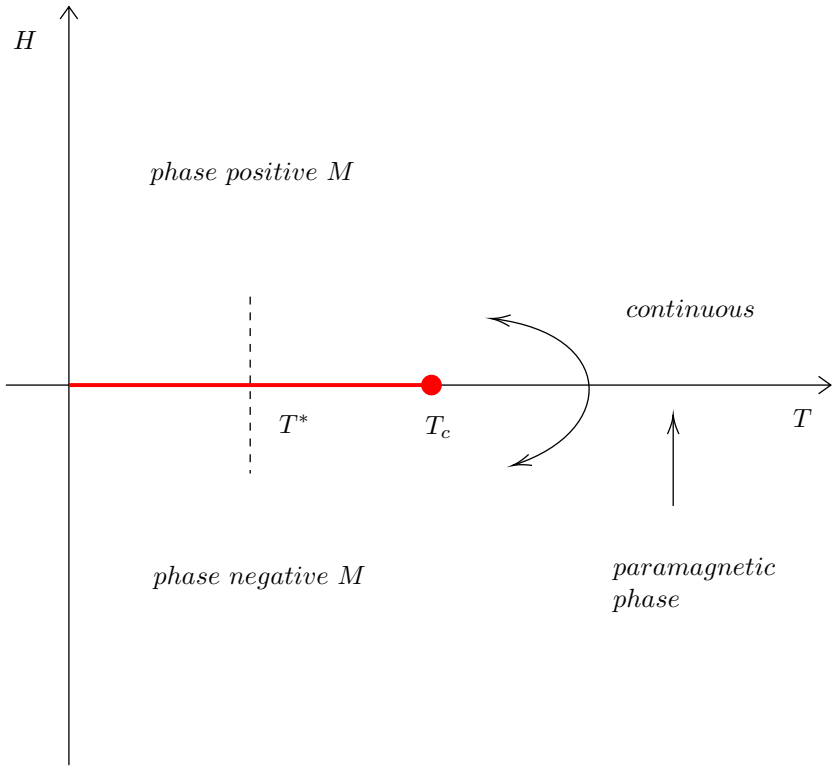


Figure 2.8: Phase diagram for a magnetic system in (T, H) space. A line of first-order transitions at zero field ends in a critical point at a temperature T_c.

The magnetization M has a jump at $H = 0$ for temperatures lower than the critical one; in this case since $M = -\frac{\partial F}{\partial H}$ we see that the first derivative of the free energy F with respect to H has a jump discontinuity. For instance, consider Figure 2.9. At the critical point the magnetization would pass through zero.

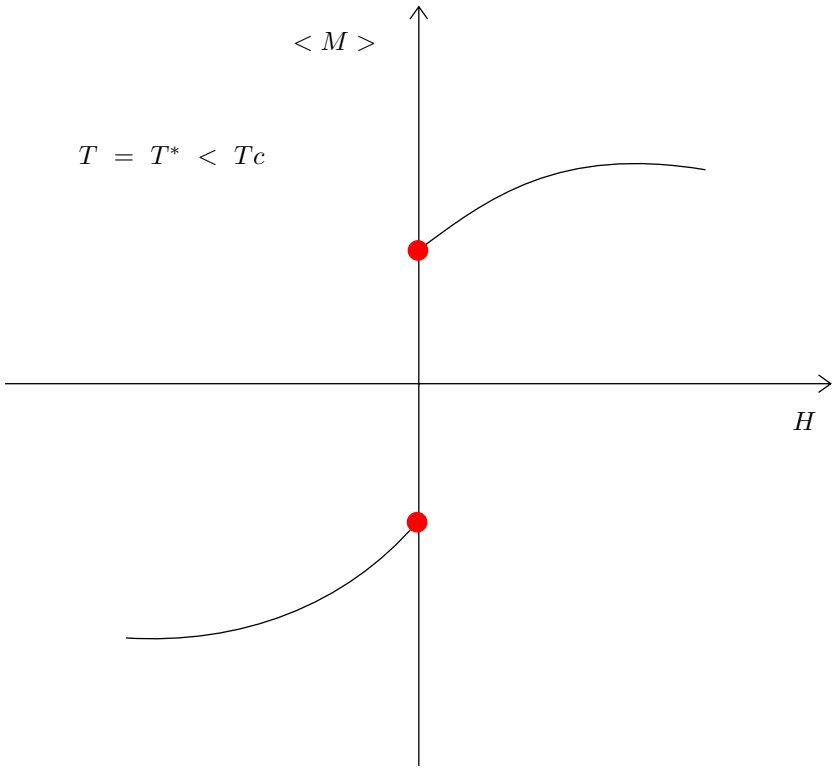


Figure 2.9: Plot of the Magnetization for $T = T^* < T_c$.

2.3 Second order phase transition

The transitions are classified in the first order transition and continuous transition. If the first derivatives are continuous, but second derivatives are discontinuous, or infinite, the transition will be described as higher order, *continuous* or critical. This is different from the previous situation, in which we had a jump for the first order derivative of a thermodynamic potential. Some examples are illustrated in Figure 2.10.

Let us suppose that

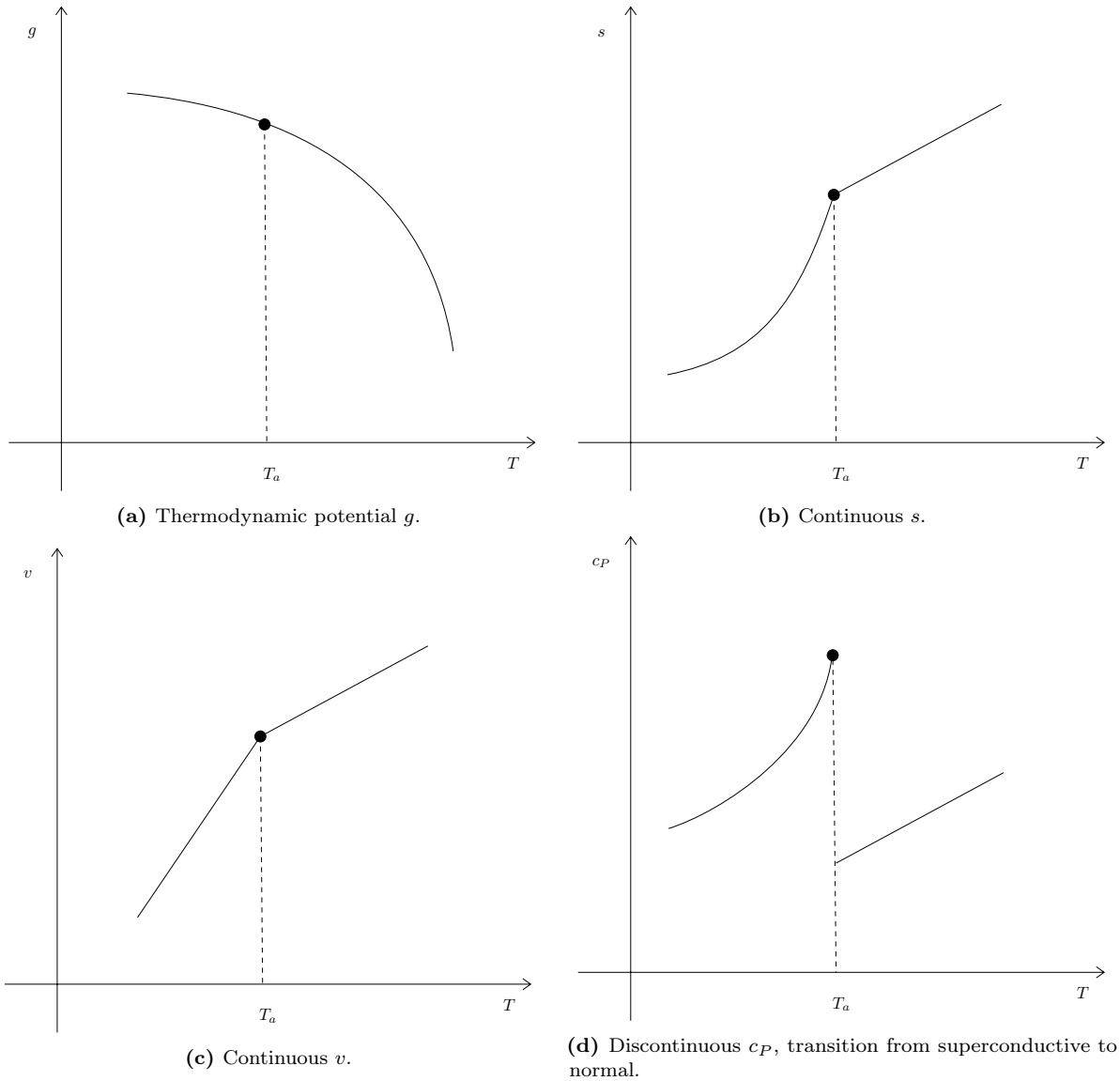
$$\left(\frac{\partial g}{\partial T}\right)_P = -s \tag{2.4}$$

$$\left(\frac{\partial g}{\partial P}\right)_T = v \tag{2.5}$$

are continuous. We suppose also that

$$\left(\frac{\partial^2 g}{\partial T \partial P}\right) = \left(\frac{\partial v}{\partial T}\right)_P = v\alpha_p \tag{2.6}$$

is discontinuous. An example is *superconductivity*.



(a) Thermodynamic potential g.

(b) Continuous s.

(c) Continuous v.

(d) Discontinuous c_P, transition from superconductive to normal.

Figure 2.10: Example of a second order phase transition.

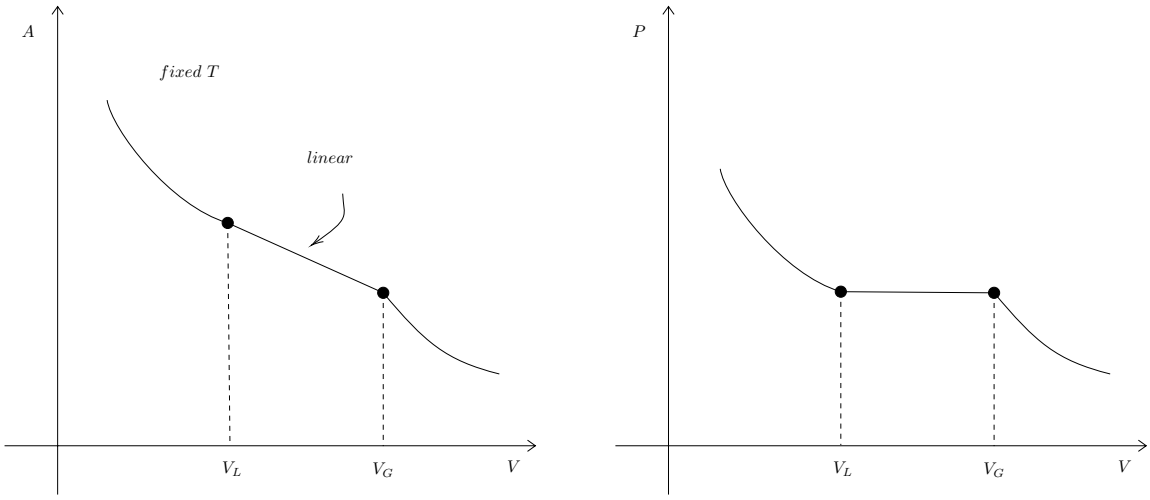
If we look for example at the specific heat c_P in Figure 2.10d, it represent the transition from superconducting.

The critical point is special because there is not a jump, so we can go continuously from gas to liquid. The response function when we plot this point shows that the specific heat diverges.

The *superfluid transition* is a transition where the second derivative of the thermodynamic potential diverges. There are many phase transitions that can be classified in different ways.

Remark. Note that at the coexistence line we increase V, but the pressure remains constant. At the coexistence line we see bubbles. It is the density that is changing locally, the bubbles becomes bigger and bigger and at the V_G, becomes a liquid.

2.3.1 Helmholtz free-energy



(a) (V, A) projection for fixed T.

(b) (P, V) projection.

Figure 2.11: Helmholtz free-energy and phase transition.

Consider $A = A(T, V, N)$, here P is replaced by V which has the derivative discontinuous at the first order transition. Moreover, $P > 0$ implies $\partial A/\partial V < 0$ and

$$k_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = -\frac{1}{V}\left(\frac{\partial P}{\partial V}\right)_T = \frac{1}{V}\left(\frac{\partial^2 A}{\partial V^2}\right)_T > 0 \tag{2.7}$$

so, A is an overall convex function of V. The behaviour of A when there is a first order phase transition is as in Figure 2.11a. The linear sector becomes an horizontal one in the $P = -(\partial A/\partial V)_T = P(V)$ curve (Figure 2.11b).

2.4 Thermodynamic of phase coexistence

2.4.1 Lever Rule

The lever rule is a rule used to determine the mole fraction of each phase of a binary equilibrium phase diagram. For instance, it can be used to determine the fraction of liquid and solid phases for a given binary composition and temperature that is between the liquid and solid line.

In an alloy or a mixture with two phases, α and β, which themselves contain two elements, A and B, the lever rule states that the mass fraction of the α phase is

$$w^\alpha = \frac{w_B - w_B^\beta}{w_B^\alpha - w_B^\beta} \tag{2.8}$$

where

- w_B^α: is the mass fraction of element B in the α phase.

- w_B^β: is the mass fraction of element B in the β phase.
- w_B: is the mass fraction of element B in the entire alloy or mixture.

Example 9

Consider Figure 2.12; at all points between A and B the system is a mixture of gas and liquid. Points D has global density $\rho_D = \rho_A + \rho_B$ and therefore $v_D = \frac{1}{\rho_D}, v_A = \frac{1}{\rho_A}, v_B = \frac{1}{\rho_B}$ which implies:

$$v_D = \frac{N_A}{N} v_A + \frac{N_B}{N} v_B = x_A v_A + x_B v_B$$

Since $x_A + x_B = 1$ we have $(x_A + x_B)v_D = x_A v_A + x_B v_B$ and finally by rearranging, one finds the *Lever Rule*. It shows that the relative concentration of the liquid-gas mixture changes with V:

$$\frac{x_A}{x_B} = \frac{v_B - v_D}{v_D - v_A}$$

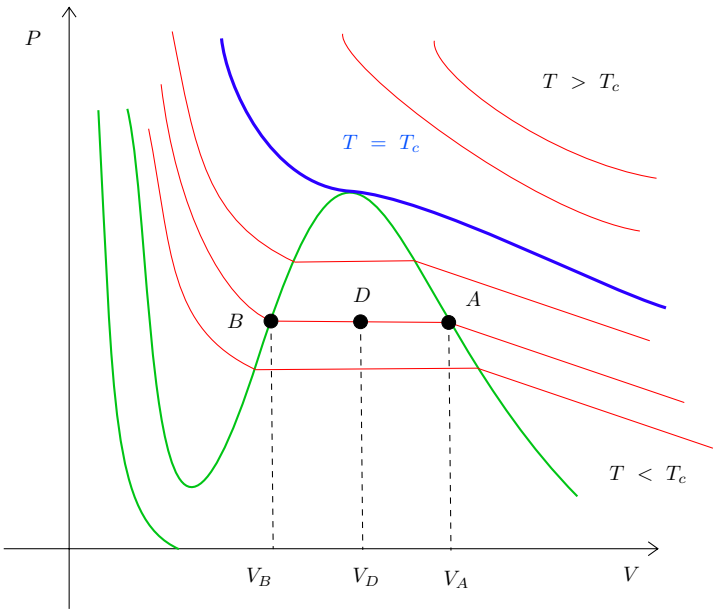


Figure 2.12: (V, P) projection. In the region between A and B the gas and the liquid phase coexist by keeping the pressure constant.

2.4.2 Phase coexistence (one component system)

Consider a (P, V, T) system as a mixture of two species $(1, 2)$ at temperature T_1, T_2, pressure P_1, P_2 and chemical potentials μ_1, μ_2. The equilibrium condition is given by the maximum of the total entropy $S = S_1 + S_2$ and gives the conditions

$$T_1 = T_2, \quad P_1 = P_2, \quad \mu_1 = \mu_2 \tag{2.9}$$

this is the *coexistence condition* of the two phases.

In terms of the Gibbs potential $G = U - TS + PV$, where U is given by the Euler equation $U = TS - PV + \mu_1 N_1 + \mu_2 N_2$, the Gibbs per mole is

$$g_1(T, P) \equiv \frac{G_1}{N_1} = \mu_1 \tag{2.10}$$

$$g_2(T, P) \equiv \frac{G_2}{N_2} = \mu_2 \tag{2.11}$$

Therefore, on the coexistence line it should hold the relation

$$g_1(T, P) = g_2(T, P) \tag{2.12}$$

2.4.3 Clausius-Clapeyron equation

The coexistence curves, as the one illustrated in Figure 2.13, are less arbitrary than is immediately evident; the slope $\mathrm{d}P\,/\,\mathrm{d}T$ of a coexistence curve is fully determined by the properties of the two coexisting phases.

The slope of a coexistence curve is of direct physical interest. Consider cubes of ice at equilibrium in a glass of water. Given the ambient pressure, the temperature of the mixed system is determined by the liquid-solid coexistence curve of water; if the temperature were not on the coexistence curve some ice would melt, or some liquid would freeze, until the temperature would again lie on the coexistence curve (or one phases would become depleted). If the ambient pressure were to decrease perhaps, by virtue of a change in altitude, then the temperature of the glass of water would appropriately adjust to a new point on the coexistence curve. If ΔP were the change in pressure, then the change in temperature would be $\Delta T = \Delta P/(\mathrm{d}P\,/\,\mathrm{d}T)_{coex}$, where the derivative in the denominator is the slope of the coexistence curve.

Remark. Ice skating presents another interesting example. The pressure applied to the ice directly beneath the blade of the skate shifts the ice across the solid-liquid coexistence curve, providing a lubricating film of liquid on which the skate slides. The possibility of ice skating depends on the negative slope of the liquid-solid coexistence curve of water.

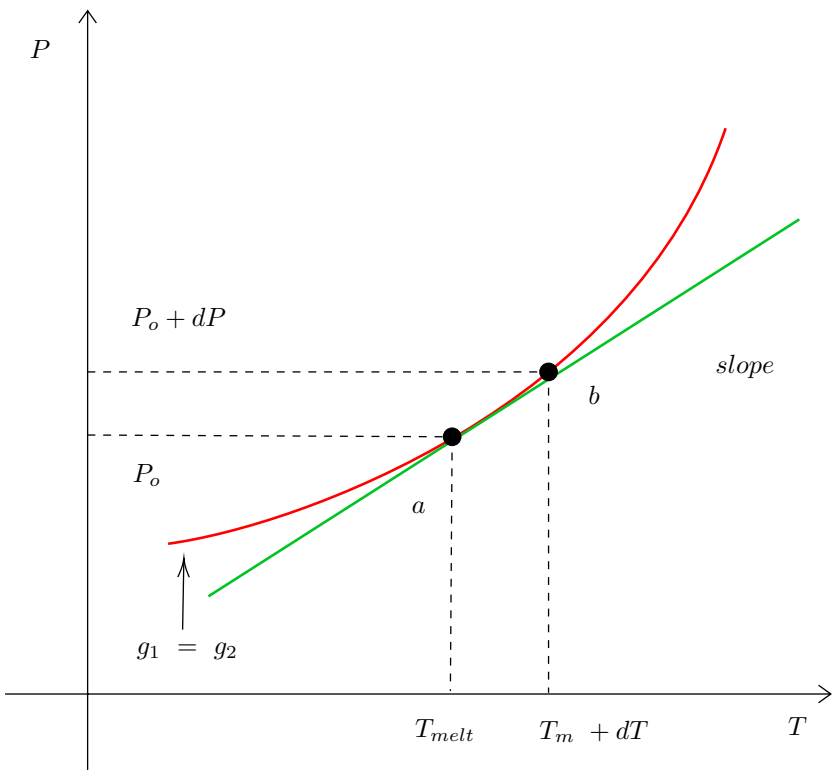


Figure 2.13: (T, P) projection. The coexistence line is represented in red, while in green the slope between the two points a and b.

Now, suppose to know the position on the coexistence line (for example the melt temperature T_m at the atmospheric pressure P_0). Is it possible to find other points on the curve? For example T_m at lower or higher pressure?

The answer is yes for small deviations of T and P from a. The idea is to compute the slope of the tangent of the coexistence curve, i.e. $(\mathrm{d}P\,/\,\mathrm{d}T)$. This is given by the Clausius-Clapeyron equation. Both at a and b the two phases 1 and 2 coexist. This means that at the coexistence line

$$\begin{cases} g_1^{(a)} = g_2^{(a)} \\ g_1^{(b)} = g_2^{(b)} \end{cases} \tag{2.13}$$

Hence, if a and b are very close:

$$\begin{cases} \mathrm{d}g_1 = g_1^{(b)} - g_1^{(a)} \\ \mathrm{d}g_2 = g_2^{(b)} - g_2^{(a)} \end{cases} \tag{2.14}$$

Therefore, the *starting point* for *Clausius-Clapeyron* is

$$\Rightarrow \mathrm{d}g_1 = \mathrm{d}g_2 \tag{2.15}$$

From the molar version of the Gibbs-Duhem relation, we have

$$\begin{cases} \mathrm{d}g_1 = -s_1\,\mathrm{d}T + v_1\,\mathrm{d}P = \mathrm{d}\mu_1 \\ \mathrm{d}g_2 = -s_2\,\mathrm{d}T + v_2\,\mathrm{d}P = \mathrm{d}\mu_2 \end{cases} \tag{2.16}$$

taking the difference, one obtains

$$-(s_2 - s_1)\,\mathrm{d}T + (v_2 - v_1)\,\mathrm{d}P = 0$$

The slope is called **Clausius-Clapeyron equation**:

$$\boxed{\left(\frac{\mathrm{d}P}{\mathrm{d}T}\right)_{coex} = \frac{(s_2 - s_1)}{(v_2 - v_1)} = \frac{\Delta s}{\Delta v}} \tag{2.17}$$

Remark. Since $(\mathrm{d}P\,/\,\mathrm{d}T)_{coex}$ is finite, the equation explains why a first order transition is characterised by discontinuous changes in entropy and volume (or density). ΔS gives the latent heat L_{12} [1]:

$$L_{12} = T\Delta s \tag{2.18}$$

whence, the Clapeyron equation is

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{L_{12}}{T\Delta v} \tag{2.19}$$

2.4.4 Application of C-C equation to the liquid-gas coexistence line

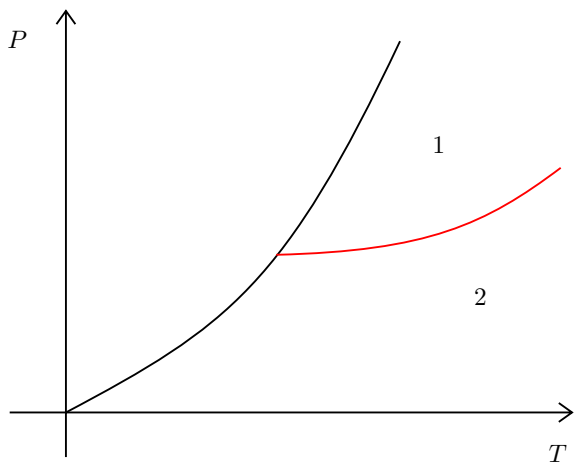


Figure 2.14: (T, P) projection. Region 1: liquid. Region 2: gas. The lines represent the combinations of pressures and temperatures at which two phases can exist in equilibrium.

Now, we go from gas (region 2) to liquid (region 1), we have:

$$\left(\frac{\mathrm{d}P}{\mathrm{d}T}\right)_{coex} = \frac{s_2 - s_1}{v_2 - v_1}$$

[1] The latent heat of fusion is the quantity of heat required to melt one mole of solid.

The Clapeyron equation embodies the *Le Chatelier principle*[2]. Consider a liquid-gas transition (the coexistence curves, from which the inequality sign is derived, are shown in Figure 2.14):

$$\left(\frac{\mathrm{d}P}{\mathrm{d}T}\right)_{coex} > 0 \quad \Rightarrow \frac{s_2 - s_1}{v_2 - v_1} > 0$$

and since $v_2 > v_1$, we have $s_2 > s_1$. The gas has more entropy as it should be. The slope of the phase curve is positive, then an increase in pressure at constant temperature tends to drive the system to the more dense (solid) phase, and an increase in temperature tends to drive the system to the more entropic (liquid) phase.

When going from a low-temperature phase to a high-temperature phase entropy always increases $\Delta S > 0$, because $c_P \equiv T(\partial S/\partial T)_P > 0$.

The sign of ΔV is more uncertain though. To see this point, let us consider the C-C equation at the solid-liquid (now solid is region 1 and liquid region 2) coexistence curve. At the melt temperature:

$$\left(\frac{\mathrm{d}P}{\mathrm{d}T}\right)_{coex} = \frac{\delta Q_{melt}}{T_{melt}\Delta v_{melt}}, \qquad \delta Q_{melt} = Q_{liq} - Q_{solid} > 0$$

In general, $\Delta v_m = v_{liq} - v_{solid} > 0$ which implies $(\mathrm{d}P\,/\,\mathrm{d}T)_{coex} > 0$. There are cases, however, where $\Delta v_m = v_{liq} - v_{solid} < 0$ because $\rho_{liq} > \rho_{solid}$ (for instance the H_20, or also Silicon and Germanium). The paradigmatic example is the freezing of water where $v_{ice} > v_{liq}$ since ice is less dense than liquid water at the coxistence ($0 < T < 4$). This implies that $\mathrm{d}P\,/\,\mathrm{d}T < 0$.

Example 10: Melting point on Everest

Consider $T = 237K$ and $P = P_0$. If we suppose that

$$\delta Q_m = 6.01kJ/mol, \quad \Delta v = -1.7cm^3/mol$$

we have

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{\delta Q_m}{T\Delta v} = \frac{6.0110^3 J/mol}{273 \cdot (-1.7cm^3/mol)} = -1.29 \cdot 10^4 J/m^3 = -1.29bar/K$$

$$\Rightarrow \Delta T = \frac{\Delta P}{(-1.29Pa/K)} = \frac{(P_0 - P_{\text{Everest}})}{(-1.29Pa/K)} = \frac{(1-0.36)atm}{(-1.29Pa/K)} = -0.5°\text{C}$$

$$\Rightarrow T_m(\text{Everest}) = T_m(P_0) + 0.5°\text{C}$$

Example 11: Boiling point on Everest

Let us consider

$$P_{\text{Everest}} = 0.36atm, \quad \rho(T = 100°\text{C}) = 0.598kg/m^3, \quad L_{gl} = 2.257 \cdot 10^3 J/g$$

The density of the vapour (gas) is about 1000 less than water (liquid), it implies that:

$$\Delta V = V_g - V_l \approx V_g = \frac{1}{\rho_g}$$

We have:

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{L_{ge}}{T\Delta V} = \frac{L_{ge}\rho_g}{T} = \frac{2.25 \cdot 10^3 J/g \cdot 0.593kg/m^3}{373K} = \frac{3.6}{K}\frac{10^3 J}{g}\frac{kg}{m^3} = 3.6 \cdot 10^3 \frac{Pa}{K}$$

$$\Rightarrow \Delta T \approx \Delta P/(3.610^3 Pa/K) = 18°\text{C}$$

$$\Rightarrow T_0 - T_{\text{Everest}} = 18°\text{C} \quad \Rightarrow T_{\text{Everest}} \approx 80°\text{C}$$

2.5 Order parameter of a phase transition

An order parameter is a measure of the degree of order across the boundaries in a phase transition system. In particular, *order parameters* are macroscopic observable that are equal to zero above the

[2]"When a settled system is disturbed, it will adjust to diminish the change that has been made to it".

critical temperature, and different from zero below:

$$O_p = \begin{cases} \neq 0 & T < T_c \\ = 0 & T \to T_c^- \end{cases} \tag{2.20}$$

When a phase transition implies a breaking of a phase symmetry, the order parameter is related to this symmetry. Therefore, the order parameter reflects the symmetry of the system. Recall that, at T_c the system has a symmetry broken.

For instance, consider the densities of liquid and gas and the related order parameter of the gas-liquid transition $\Delta\rho = \rho_l - \rho_g$, that is $\neq 0$ for $T \neq T_c$ but $\to 0$ when $T \to T_c$ (see Figure 2.15).

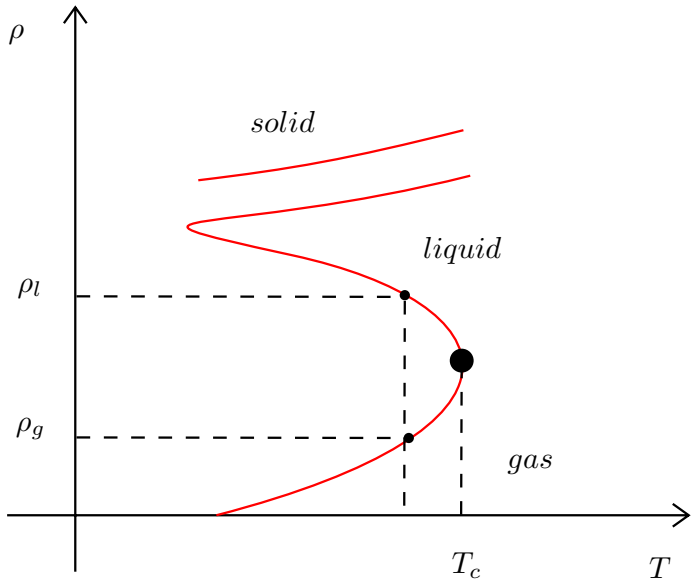


Figure 2.15: (T, ρ) projection of the (P, V, T) system, where $\rho = N/V$.

Remark. Note that $\rho = \frac{N}{V} = \frac{1}{v}$ hence either N or V varies.

In Figure 2.16 is shown the behaviour for a ferromagnetic system. We have

$$H = 0 \Rightarrow \begin{cases} M \neq 0 & T < T_c \\ M \to 0 & T \to T_c^- \end{cases}$$

Clearly $M \neq 0$ if $H \neq 0$. Recall that M is the order parameter of the paramagnetic-ferromagnetic phase transition.

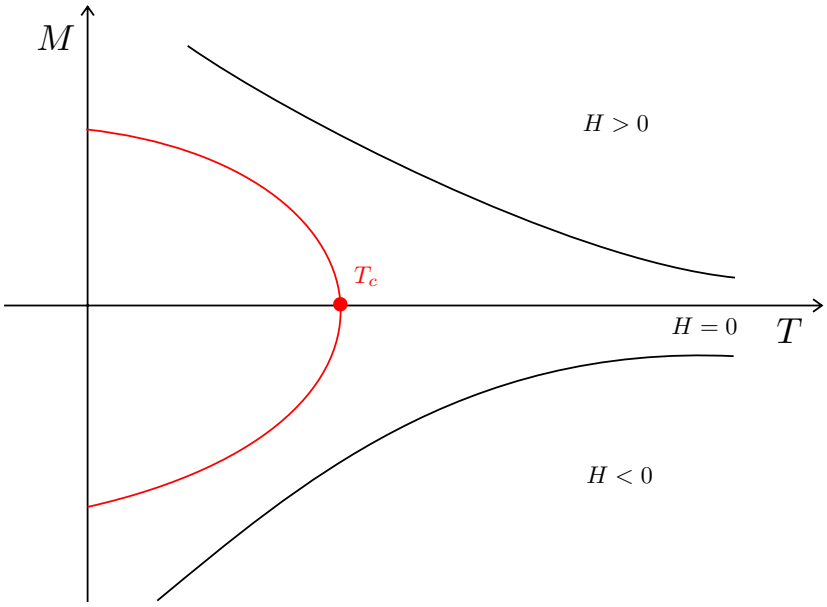


Figure 2.16: Magnetization of a ferromagnet. In red: zero-field magnetization. Below the critical temperature there is a spontaneous magnetization.

It can be either a scalar ($\Delta\rho$), a vector ($\vec{M}$) or even tensor ($Q_{\alpha\beta}$).

Variable conjugate to O_P

- *Ferromagnetic system*: $\vec{\mathbf{M}} \to \vec{\mathbf{H}}$ (magnetic field).
- *Ferroelectric*: $\vec{\mathbf{P}} \to \vec{\mathbf{E}}$ (electric field).
- *Liquid crystals*: $Q_{\alpha\beta} \to \vec{\mathbf{E}}, \vec{\mathbf{H}}$.
- *Fluid*: $V \to P$ (pressure), or $\rho \to \mu$.

2.6 Classification of the phase transitions

2.6.1 Thermodynamic classification

Thermodynamically, one can distinguish two kinds of phase transitions:

1. Ones who develop latent heat.
2. Ones who do not develop latent heat. The entropy changes continuously at the transition.

2.6.2 Eherenfest classification

The *Eherenfest classification* is based on the behaviour of the derivatives of the thermodynamic potentials.

A phase transition is of order n if all the $(n-1)$ derivatives are continuous and the nth derivative displays a finite discontinuity.

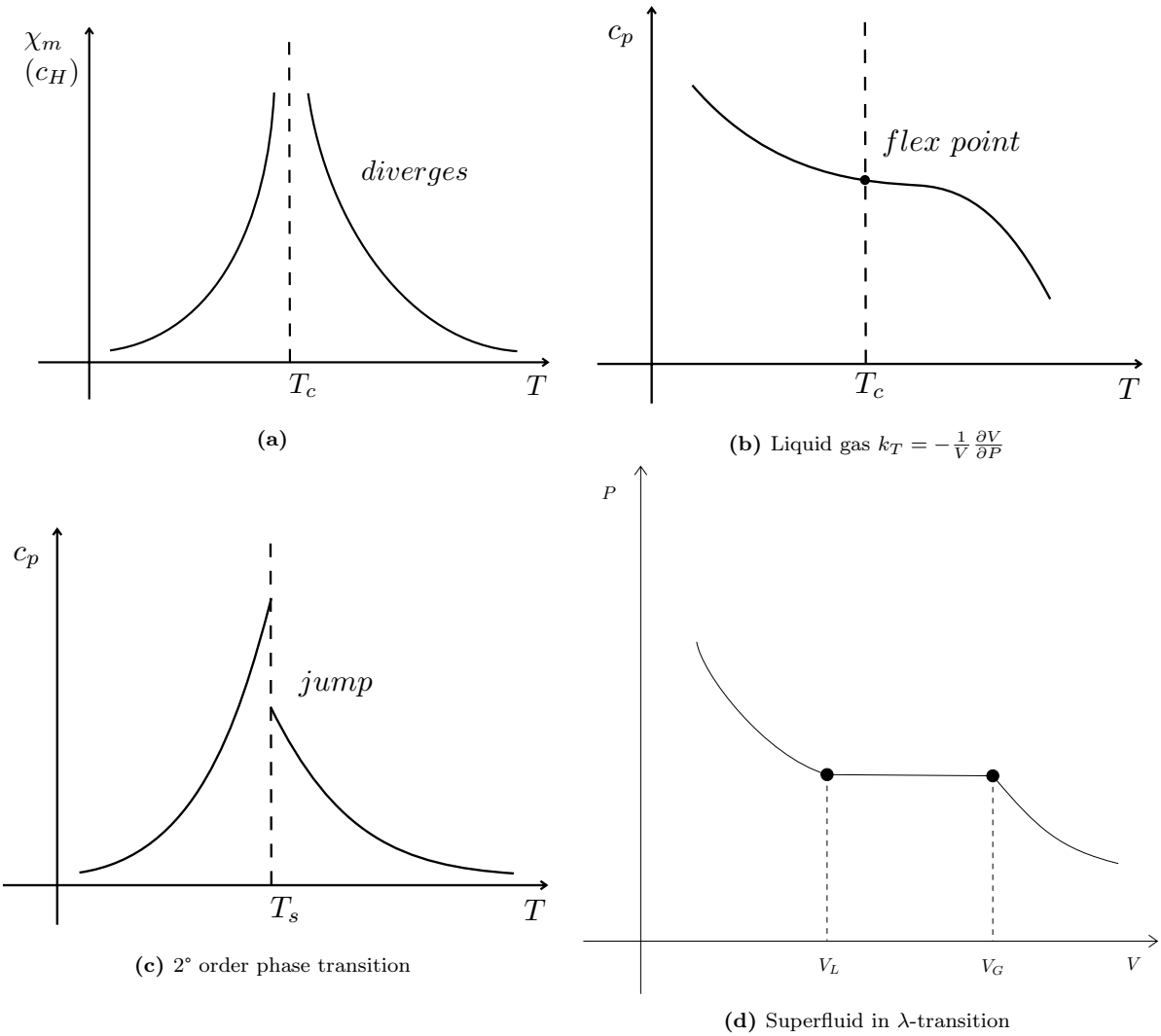
Example 12

For instance, a first order transition in which $S = -(\partial G/\partial T)_P$ has finite discontinuity.

Remark. There are first order transitions where S is continuous (no latent heat), but ρ is discontinuous ($v = (\partial G/\partial P)_T$).

Example 13

Second order transition. The specific heat displays a finite jump, see Figure 2.17c in the conductor-superconductor transition.
Another example is a second order transition but with divergence. Consider the fluid-superfluid transition (or λ transition) of the He_4 (Figure 2.17d).



(a)

(b) Liquid gas $k_T = -\frac{1}{V}\frac{\partial V}{\partial P}$

(c) 2° order phase transition

(d) Superfluid in λ-transition

Figure 2.17: Plots of response functions.

Remark. λ transition: a second-order or higher-order transition, in which the heat capacity shows either a discontinuity (second-order) or a vertex (higher-order) at the transition temperature. It is so named because the shape of the specific heat versus temperature curve resembles the Greek letter λ.

2.6.3 Modern classification

A phase transition is of the **first order** if exists a finite discontinuity in either one or more partial derivatives of the thermodynamic potentials. Instead, if the first derivatives are all continuous, but the second are either discontinuous, or infinite, one talks of **continuous transitions**. A critical point is a continuous transition.

2.7 Critical exponents

At the critical point response functions may diverge. How are these divergence? In general, when you are close to T_c, there are singolarities. Now, we can ask, how the curve diverges? What is the behaviour close to the critical point? Power law, so which are the values of these critical exponents?

2.7.1 Divergence of the response functions at the critical point

While at the critical point the order parameter goes to zero continuously as $T \to T_c^-$, the response function may develop divergences.

Example 14

In a fluid system since at $T = T_c$ the curve $P = P(V)$ develops an horizontal flex , we have $k_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \to \infty$. Similarly, in a magnetic since the curve is like Figure 2.16, we have $\chi_T = \left(\frac{\partial M}{\partial H}\right)_T \underset{T\to T_c}{\to} \infty$.

2.7.2 Critical exponents definition

The notion of *critical exponent* describes the behaviour of the order parameter and the response functions in proximity of the critical point. In order to answer to these questions, let us define:

Definition 1: Critical Exponent, *or* Scale Exponent

Let us define the adimensional parameter measuring the distance from the critical point $t \equiv \frac{T-T_c}{T_c}$, the *critical exponent* λ associated to the function $F(t)$ is defined as:

$$\lambda_\pm = \lim_{t\to 0^\pm} \frac{\ln|F(t)|}{\ln|t|} \tag{2.21}$$

We note that it behaves like a power law. One can also write the *power law*:

$$F(t) \overset{t\to 0^\pm}{\sim} |t|^{\lambda_\pm} \tag{2.22}$$

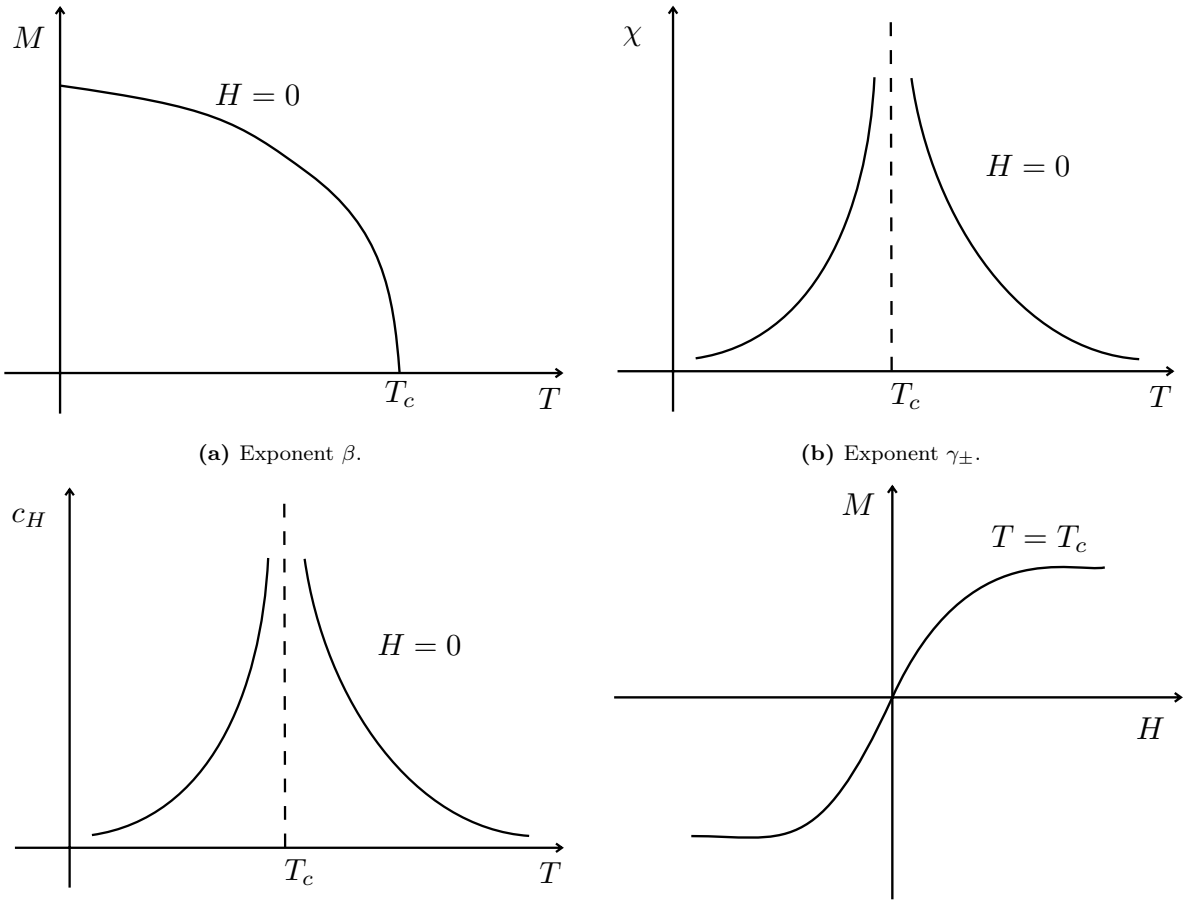
More generally, for $t \ll 1$:

$$F(t) = A|t|^{\lambda_\pm}(1 + bt^{\lambda_1} + \dots), \quad \lambda_1 > 0 \tag{2.23}$$

where all other terms are less important.

Definition 2: Thermodynamic Critical Exponents

- **Exponent $\boldsymbol{\beta}$**: tells how the order parameter goes to zero. Consider Figure 2.18a, we have $|M| \overset{t\to 0^-}{\sim} (-t)^\beta$. No sense in going from above ($t \to 0^+$) where it stays 0.
- **Exponent $\boldsymbol{\gamma_\pm}$** (susceptibility): related to the response function. Consider Figure 2.18b, we have $\chi_T \overset{t\to 0^\pm}{\sim} |t|^{-\gamma_\pm}$. In principle, the value of γ can depend on the sign of t i.e. $\gamma^+ \neq \gamma^-$, but they are the same in reality and we have $\gamma^+ = \gamma^- = \gamma$.
- **Exponent $\boldsymbol{\alpha_\pm}$**: how specific heat diverges (second order derivative in respect of T). For instance see Figure 2.18c, we have $c_H \sim |t|^{-\alpha_\pm}$.
- **Exponent $\boldsymbol{\delta}$**: in this case one consider the isotherm $T = T_c$ and look for the behaviour of M at the critical point at small H (or viceversa). The result is $M \sim H^{1/\delta}$. In Figure 2.18d, $H \sim |M|^\delta \mathrm{sign}(M)$.



(a) Exponent β.

(b) Exponent $\gamma_\pm$.

(c) Exponent $\alpha_\pm$.

(d) Exponent δ.

Figure 2.18

Zero-field specific heat	$C_H \sim \lvert t\rvert^{-\alpha}$
Zero-field magnetization	$M \sim (-t)^\beta$
Zero-field isothermal susceptibility	$\chi_T \sim \lvert t\rvert^{-\gamma}$
Critical isotherm ($t = 0$)	$H \sim \lvert M\rvert^\delta \mathrm{sign}(M)$
Correlation length	$\xi \sim \lvert t\rvert^{-\nu}$
Pair correlation function at T_c	$G(\vec{\mathbf{r}}) \sim \frac{1}{r^{d-2+\eta}}$

Table 2.1: Definitions of the most commonly used critical exponents for a magnetic system

Specific heat at constant volume V_c	$C_V \sim \lvert t\rvert^{-\alpha}$
Liquid-gas density difference	$(\rho_l - \rho_g) \sim (-t)^\beta$
Isothermal compressibility	$k_T \sim \lvert t\rvert^{-\gamma}$
Critical isotherm ($t = 0$)	$P - P_c \sim \lvert\rho_l - \rho_g\rvert^\delta \mathrm{sign}(\rho_l - \rho_g)$
Correlation length	$\xi \sim \lvert t\rvert^{-\nu}$
Pair correlation function at T_c	$G(\vec{\mathbf{r}}) \sim \frac{1}{r^{d-2+\eta}}$

Table 2.2: Definitions of the most commonly used critical exponents for a fluid system

Remark. In compiling Table 2.1 and 2.2 we have made the as yet totally unjustified assumption that the critical exponent associated with a given thermodynamic variable is the same as $T \to T_c$ from above or below.

Critical exponents depend on very few parameters and are not system-specific:

1. Symmetries of the System
2. Dimensions of the System
3. Range of Interaction

T_c instead does depend on specific chemistry, is not as universal as critical exponent.

2.7.3 Law of the corresponding states (Experimental estimates of order parameters)

The system displays correlation at very long distance, these goes to the size of the system when $T \to T_c$. We are talking about long range correlation. The *correlation function* is $\xi \sim t^{-\nu}$. For instance, consider a polymer as in Figure 2.19a.

Having defined the critical exponents, we need to justify why they are interesting and why they are more interesting than the critical temperature T_c itself. It turns out that, whereas T_c depends sensitively on the details of the interatomic interactions, the critical exponents are to a large degree *universal* depending only on a few fundamental parameters.

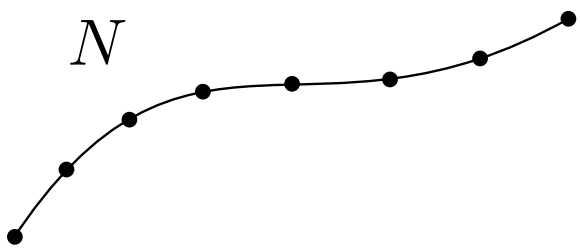
To summurize, the critical exponents are more interesting than T_c since their values do not depend on microscopic details, but only on few parameters such as the space dimension d and the symmetry of the system.

One of the first experimental evidence of this universality was given by the work of Guggenheim on the coexistence curves of g different fluids: A, Kn, χ_e, Ne, N_2, CO_2 and O_2. By plotting T/T_c versus ρ/ρ_c (Figure 2.19b) he found that all the data collapse on the same curve, i.e. different sets of data fit the same function. Moreover for $t \to 0$:

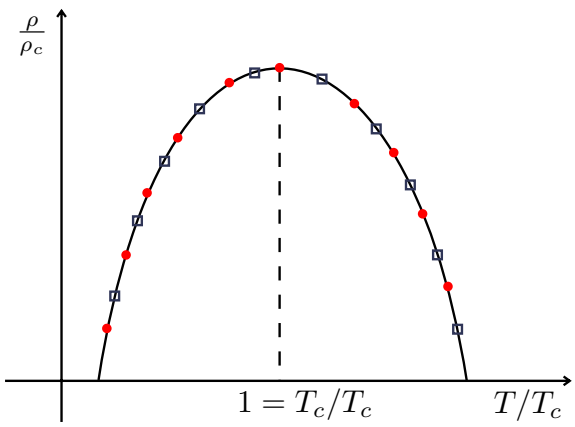
$$(\rho_l - \rho_c) \sim (-t)^\beta$$

and $\beta \sim 1/3 \approx 0.335$. Therefore, close to the critical point all the data lie on the same curve and hence can be described by the same exponent β. A further test of universality is to compare this value to that obtained for a phase transition in a completely different system with a scalar order parameter. For instance, if we do the same for a string ferromagnetic the result is $\beta = 1/3$ too.

Remark. The law of corresponding states gives a universal liquid-gas coexistence curve.



(a) N-Polymer.



(b) Coexistence curve of different fluids plotted in reduced variables.

Figure 2.19

2.7.4 Thermodynamic inequalities between critical exponents

It is possible to obtain several rigorous inequalities between the critical exponents. The easiest to prove is due to Rushbrooke.

Rushbrocke inequality

It follows from the well known thermodynamic relation between the specific heats at constant field and constant magnetization. Remember the relation between response functions: For a fluid:

$$k_T(c_p - c_v) = Tv\alpha^2 = Tv\frac{1}{v^2}\left(\frac{\partial v}{\partial T}\right)_P^2 = T\frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_P^2$$

For magnetic systems one has

$$\chi_T(c_H - c_M) = T\underbrace{\left(\frac{\partial M}{\partial T}\right)_H^2}_{\geq 0}$$

From thermodynamic stability we have $c_M \geq 0, c_H \geq 0, \chi_T \geq 0$. Hence, from the previous relation we have

$$c_H = \frac{T}{\chi_T}\left(\frac{\partial M}{\partial T}\right)_H^2 + \underbrace{c_M}_{\geq 0}$$

which implies

$$c_H \geq \frac{T}{\chi_T}\left(\frac{\partial M}{\partial T}\right)_H^2 \tag{2.24}$$

On the other hand, for $T \to T_c^-$ $(t \to 0^-)$ and $H = 0$ (zero field) we have

$$\begin{cases} c_H \sim (-t)^{-\alpha} \\ \chi_T \sim (-t)^{-\gamma} \\ M \sim (-t)^{\beta} \end{cases}$$

that implies

$$\left(\frac{\partial M}{\partial T}\right)_{H=0} \sim (-t)^{\beta-1}$$

Since the inequality (2.24) is valid for all temperature T, it follows that can only be obeyed if

$$B(T_c - T)^{-\alpha} \geq B'T\frac{[(T_c - T)^{\beta-1}]^2}{(T_c - T)^{-\gamma}}$$

with $B, B' > 0$. Take the limit $T \to T_c^-$, we have:

$$\lim_{T\to T_c^-}(T_c - T)^{2-\alpha-2\beta-\gamma} \geq \frac{B'T}{B} > 0$$

Since the left hand side must be strictly greater than zero, we have the *RushBrook inequality*:

$$\alpha + 2\beta + \gamma \geq 2 \tag{2.25}$$

Griffith inequality

The *Griffith inequality* is obtained from the convexity property (in T and V) of the Helmolds free energy and from $A \sim t^{2-\alpha}$ (as $t \to 0^-$ deduced from $c_P = t^{-\alpha}$):

$$\Rightarrow \alpha + \beta(1 + \delta) \geq 2 \tag{2.26}$$

We have introduced two very new ideas, universality and inequalities between the critical exponents, which appear to hold as equalities.

In the intervening chapters, we look at models of systems which undergo phase transitions and how to calculate their critical exponents and other properties.

Chapter 3

Recall of statistical mechanics and theory of ensembles

3.1 Statistical ensembles

Statistical mechanics roughly speaking was born as a sort of theory from microscopic trying to compute the macroscopic length using thermodynamics. The problem is going from the countinuous problems to the macroscopic problems. In origin was statistical mechanics of equilibrium system. Each microstate with a given energy fixed, will have the same probability, this is the equal probability statement.

3.1.1 Microcanonical ensemble

In general, if we consider a system with N, V (number of particles and volume) fixed and also the total energy E fixed, we call $\Omega(E, V, N)$ the number of microstate with total energy E , volume V and number of particles N.

If the system is *isolated* and in *equilibrium* the rule of **equal probability** of the microstates holds:

If the system is isolated and in equilibrium with energy E it visits each microstate consistent with energy E with equal probability.

Another way to say is: the system spends the same amount of time in each of the $\Omega(E, V, N)$ microstates.

Therefore, we call a single configuration of a given microstate $\mathcal{C}$. A configuration is just when you have the spatial part, because momentum can be obtained by integrating. Let us suppose to compute the probability of a given configuration $\mathcal{C}$, $P_{\mathcal{C}}$; because of equal probability we have:

$$P_{\mathcal{C}} = \frac{1}{\Omega(E, V, N)} \tag{3.1}$$

Now, let us now consider two subsystem 1 and 2 that can exchange energy, volume and/or particles. The number of microstates, of the combined system, of total energy $E_T = E_1 + E_2$, total volume $V_T = V_1 + V_2$ and $N_T = N_1 + N_2$ is given by:

$$\Omega(E_T, V_T, N_T) = \sum_{E_1, V_1, N_1} \Omega_1(E_1, V_1, N_1)\Omega_2(E_T - E_1, V_T - V_1, N_T - N_1) \tag{3.2}$$

One can show that, in the thermodynamic limit, $\Omega(E_T, V_T, N_T)$ is strongly peaked around a given point (E_1^*, V_1^*, N_1^*) and the fluctuations around this value are rare and small. Writing $\Omega(E_T, V_T, N_T)$ as

$$\Omega(E_T, V_T, N_T) \propto e^{\frac{S(E_T, V_T, N_T)}{k_B}} = \sum_{E_1, V_1, N_1} \exp\left[\frac{1}{k_B}(S_1(E_1, V_1, N_1) + S_2(E_2, V_2, N_2))\right] \tag{3.3}$$

(the proportionality becomes from the Boltzmann definition of entropy).

The values (E_1^*, V_1^*, N_1^*) are obtained by the max entropy condition that can be written as

$$\frac{\mathrm{d}\ln\Omega_1}{\mathrm{d}E_1} = \frac{\mathrm{d}\ln\Omega_2}{\mathrm{d}E_2} \Rightarrow T_1 = T_2 \tag{3.4a}$$

$$\frac{\mathrm{d}\ln\Omega_1}{\mathrm{d}V_1} = \frac{\mathrm{d}\ln\Omega_2}{\mathrm{d}V_2} \Rightarrow P_1 = P_2 \tag{3.4b}$$

$$\frac{\mathrm{d}\ln\Omega_1}{\mathrm{d}N_1} = \frac{\mathrm{d}\ln\Omega_2}{\mathrm{d}N_2} \Rightarrow \mu_1 = \mu_2 \tag{3.4c}$$

We next consider these properties to the case in which 1 is the system we want to study and 2 is a much larger system than 1 (a bath). This setup will bring us to the *canonical ensemble*.

3.2 The canonical ensemble

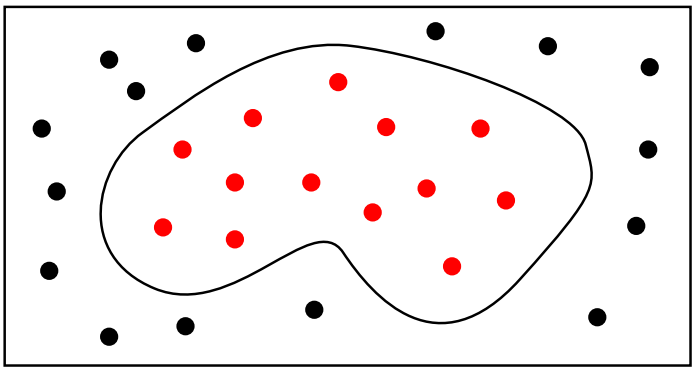


Figure 3.1: Isolated system. There are two subsystems, S constituted by red points and B constituted by the black one.

Let us consider an isolated system made by two subsystems, one S and one much larger, B, that we call *thermal bath* (Figure 3.1). The total number of particles is given by $N_T = N_B + N_S$ with $N_B \gg N_S \gg 1$ (they are both large but B is much larger than S), where N_B are the particles in the thermal bath and N_S the particle of the system.

Let E_T be the energy of the composite system. The two subsystems can exchange energy but the whole system has constant energy E_T. Therefore, let the energy to be free to fluctuate in time at fixed temperature T_B (isothermal ensembles). Note that V_S, N_S, V_B, N_B are fixed (no exchange of volume and particles).

For resuming, other quantities fixed are the temperature of the bath T_B , the number of the total particles of the system N_T, and also the total volume V_T. We have also $V_T = V_B + V_S$, with $V_B \gg V_S$.

The key to the canonical formalism is the determination of the probability distribution of the system among its microstates. And this problem is solved by the realization that the system plus the bath constitute a closed system, with fixed temperature, to which the principle of equal probability of microstates applies.

If one assumes that the system and the bath are *weakly coupled* (neglet interaction energy):

$$E_T = E_S + E_B = const \qquad E_B \gg E_S$$

Let $\mathcal{C}$ by the microstate of the system S, and $\mathcal{G}$ the microstate of the heat bath B. A given microstate of the isolated composite system B-S is given from a pair $(\mathcal{C}, \mathcal{G})$ of microstate $\mathcal{C} \in S$ and $\mathcal{G} \in B$. The number of microstates of the isolated system with total energy E_T and system energy E_S is given by:

$$\Omega_T(E_T, E_S) = \Omega(E_S)\Omega_B(E_T - E_S)$$

Remark. In this analysis V and N are fixed. Since E_T is fixed

$$\Omega_T(E_T) = \sum_{E_S} \Omega(E_S)\Omega_B(E_T - E_S) \tag{3.5}$$

From the principle of equal probability for microstates at equilibrium, the probability of a composed microstate $(\mathcal{C} \circ \mathcal{G})$ is given by:

$$P_{\mathcal{C}\circ\mathcal{G}} = \begin{cases} \frac{1}{\Omega_T(E_T)} & E_{\mathcal{C}} + E_{\mathcal{G}} = E_T \\ 0 & \text{otherwise} \end{cases} \tag{3.6}$$

Since we are not interested to the microstates of the heat bath

$$P_{\mathcal{C}} = \sum_{\substack{\text{all } \mathcal{G} \\ \text{such that} \\ g(E_T - E_{\mathcal{C}} - E_{\mathcal{G}})}} P_{\mathcal{C}\circ\mathcal{G}} = \sum_{\substack{\text{all } \mathcal{G} \\ \text{such that} \\ g(E_T - E_{\mathcal{C}} - E_{\mathcal{G}})}} \frac{1}{\Omega_T(E_T)} = \frac{1}{\Omega_T} \sum_{\mathcal{G}} 1 \tag{3.7}$$

The number of microstates $\mathcal{G}$ with energy $E_{\mathcal{G}} = E_T - E_{\mathcal{C}}$ is given by:

$$\Omega_B(E_{\mathcal{G}}) = \Omega_B(E_T - E_{\mathcal{C}})$$

This implies that the probability of a given configuration is related to the number of microstate of the bath:

$$\Rightarrow P_{\mathcal{C}} = \frac{\Omega_B(E_T - E_{\mathcal{C}})}{\Omega_T(E_T)} \propto \Omega_B(E_T - E_{\mathcal{C}}) \tag{3.8}$$

It is more convenient to deal with the logarithmic of $P_{\mathcal{C}}$ that is smoother

$$\Rightarrow k_B \ln \Omega_B(E_T - E_{\mathcal{C}}) = S_B \tag{3.9}$$

This is the entropy of B and is a function of N_B. Since $E_{\mathcal{C}} \ll E_B \simeq E_T$ we can expand $S_B(E_T - E_{\mathcal{C}})$ around $x_0 = E_T$ by the small amount

$$\Delta \equiv x - x_0 = (E_B) - (E_T) = -E_{\mathcal{C}}$$

$$f(E_B) = f(E_T) + \left.\frac{\mathrm{d}f}{\mathrm{d}E_B}\right|_{E_B=E_T} (E_B - E_T) + \dots$$

Therefore:

$$k_B \ln \Omega_B(E_B) = S_B(E_B) = S_B(E_T) - E_{\mathcal{C}} \left(\frac{\partial S_B}{\partial E_B}\right)_{E_B=E_T} + \frac{E_{\mathcal{C}}^2}{2} \left(\frac{\partial^2 S_B}{\partial E_B^2}\right)_{E_B=E_T} + \dots \tag{3.10}$$

To make explicit the N_B dependence, let us consider the molar version

$$S_B \to N_B s_B \qquad E_B \to N_B e_B$$

$$S_B = N_B s_B = N_B s_B(E_T) - E_{\mathcal{C}} \left(\frac{\partial s_B}{\partial e_B}\right)_{e_B=e_T} + \frac{E_{\mathcal{C}}^2}{2N_B} \left(\frac{\partial^2 s_B}{\partial e_B^2}\right)$$

Let us consider the limit in which the system size is fixed, while the one of the heat bath is going to ∞:

$$\lim_{N_B\to\infty} \frac{E_T}{N_B} = \frac{E_S + \overbrace{N_B e_B}^{E_B}}{N_B} \to e_B \tag{3.11a}$$

$$\lim_{N_B\to\infty} k_B \ln \Omega_B(E_T - E_{\mathcal{C}}) \to N_B s_B - E_{\mathcal{C}} \frac{\mathrm{d}s_B}{\mathrm{d}e_B} \tag{3.11b}$$

On the other hand,

$$\frac{\mathrm{d}s_B}{\mathrm{d}e_B} \equiv \frac{1}{T_B} = \frac{1}{T}$$

which implies

$$P_{\mathcal{C}} \propto \Omega_B(E_T - E_{\mathcal{C}}) = \exp\left(\frac{S_B}{k_B}\right) = \exp\left(\frac{N_B s_B}{k_B} - \frac{E_{\mathcal{C}}}{k_B T}\right)$$

Since the first therm does not depend on $\mathcal{C}$, it can be absorbed in the constant and what we get by expanding considering the huge number of particles

$$P_{\mathcal{C}} \propto \exp(-E_{\mathcal{C}}/k_B T) \tag{3.12}$$

Remark. Since the energy of the system fluctuates, its microstates are not anywhere equiprobable, but are visited with probability given by (3.12).

Remark. Since the bath is very large, T is the only property of the bath that affects the system. The *Boltzmann factor* is defined as:

$$\beta \equiv \frac{1}{k_B T} \tag{3.13}$$

The normalization consists in dividing by the normalization factor, that is the sum of all microstates

$$P_{\mathcal{C}} = \frac{e^{-\beta E_{\mathcal{C}}}}{\sum_{\mathcal{C}} e^{-\beta E_{\mathcal{C}}}} \tag{3.14}$$

Finally, the **canconical partition function** is defined as

$$Q(T,V,N) \equiv \sum_{\substack{\text{all } \mathcal{C} \\ \text{with } V,N \\ \text{fixed}}} \exp(-\beta E_{\mathcal{C}}) \tag{3.15}$$

Given $Q(T,V,N)$, one gets the equation that states the link between statistical mechanics and thermodynamics, which is the *Helhmoltz free energy*:

$$A(T,V,N) = -k_B T \ln Q(T,V,N) \tag{3.16}$$

that is the free energy describing the isothermal (or canonical) ensemble at fixed T, volume V and number of particles N.

Remark.

$$Q(T,V,N) = \sum_{\substack{\mathcal{C} \\ V,N\text{fixed}}} e^{-\beta E(\mathcal{C})} = \sum_E e^{-\beta E} \Omega(E,V,N)$$

What we have done is a *foliation* in energy of the space, that is a sum over the energy (keeping $\{V,N\}$ fixed):

$$Q(T,V,N) = \sum_E e^{-\beta E} \Omega(E,V,N) = \sum_E e^{-\beta E} e^{S/k_B} = \sum_E e^{-\beta(E-TS)}$$

Hence,

$$Q(T,V,N) = e^{-\beta A} \quad \Rightarrow A = -k_B T \ln Q(T,V,N)$$

We have formulated a complete algorithm for the calculation of a fundamental relation in the canonical formalism. Given a list of states of the system, and their energies $E_{\mathcal{C}}$, we calculate the partition function (3.15). The partition function is thus obtained as a function of temperature and of the parameters that determine the energy levels. The fundamental relation is (3.16) that determines the Helmholtz potential.

The probability of a configuration can be written as (3.14), that is a very useful form. Indeed, the average energy is expected to be

$$U = \sum_{\mathcal{C}} E_{\mathcal{C}} P_{\mathcal{C}} = \frac{\sum_{\mathcal{C}} E_{\mathcal{C}} e^{-\beta E_{\mathcal{C}}}}{\sum_{\mathcal{C}} e^{-\beta E_{\mathcal{C}}}} \tag{3.17}$$

or

$$U = -\frac{\partial}{\partial \beta} \ln Q \tag{3.18}$$

3.2.1 Energy fluctuations in the canonical ensemble

Despite energy in the canonical ensemble fluctuates, while in the microcanonical one is constant, this does not contradict the equivalence principle of the ensemble (in the thermodynamic limit). The reason is that the relative size of the energy fluctuation decreases in the large system limit. To see it, let us compute the average square fluctuations of E.

$$\left\langle (\delta E)^2 \right\rangle = \left\langle (E - \langle E \rangle)^2 \right\rangle = \left\langle E^2 \right\rangle - \langle E \rangle^2 \tag{3.19}$$

Remark. Remember that thermodynamic assume that the number of number of freedom is related to the number of Avogadro.

On the other hand,

$$\langle E\rangle = \sum_{\mathcal{C}} P_{\mathcal{C}} E_{\mathcal{C}} = \sum_{\mathcal{C}} E_{\mathcal{C}} \frac{e^{-\beta E_{\mathcal{C}}}}{\sum_{\mathcal{C}} e^{-\beta E_{\mathcal{C}}}} = -\left(\frac{\frac{\partial Q(T,V,N)}{\partial \beta}}{Q}\right) = -\left(\frac{\partial}{\partial \beta}\ln Q\right)_{N,V}$$

$$\left\langle E^2\right\rangle = \sum_{\mathcal{C}} P_{\mathcal{C}} E_{\mathcal{C}}^2 = \frac{\left(\frac{\partial^2 Q}{\partial \beta^2}\right)}{Q}$$

Therefore,

$$\begin{aligned}\left\langle (\delta E)^2\right\rangle = \left\langle (E-\langle E\rangle)^2\right\rangle &= \frac{1}{Q}\left(\frac{\partial^2 Q}{\partial \beta^2}\right)_{N,V} - \frac{1}{Q^2}\left(\frac{\partial Q}{\partial \beta}\right)^2_{N,V} \\ &= \left(\frac{\partial^2 \ln Q}{\partial \beta^2}\right)_{N,V} = -\left(\frac{\partial \langle E\rangle}{\partial \beta}\right)_{N,V}\end{aligned}$$

Since

$$c_V = \left(\frac{\partial E}{\partial T}\right)_{N,V} \tag{3.20}$$

we have

$$\left\langle (\delta E)^2\right\rangle = k_B T^2 c_V \tag{3.21}$$

Both c_V and $\langle E\rangle$ are extensive

$$\frac{\sqrt{\langle (\delta E)^2\rangle}}{\langle E\rangle} = \frac{\sqrt{k_B T^2 c_V}}{\langle E\rangle} \sim O\left(\frac{1}{\sqrt{N}}\right) \Rightarrow 0$$

because $N \sim 10^{23}$.

3.3 Isothermal and isobaric ensemble

Now, the system is coupled both to a thermal and a volumic bath at temperature T_B and pressure P_B. The idea is: consider the same system with the bath; the difference is that in this case the system can exchange energy but also volume (we continue to keep the temperature of the bath fixed). At this point the ensemble is *isothermal* and *isobaric*. All the assumptions done before are valid, in particular, assuming as before *weak coupling* between the degrees of freedom of the bath and those of the system

$$\begin{aligned}E_T &= E + E_B \\ V_T &= V + V_B\end{aligned}$$

We look for the partition function that describes this isothermal and isobaric ensemble. Similarly to the previous case, one can write

$$P_{\mathcal{C}} \propto \Omega_B(E_B, V_B) \propto \Omega_B(E_T - E_{\mathcal{C}}, V_T - V_{\mathcal{C}}) \propto \exp[S_B(E_T - E_{\mathcal{C}}, V_T - V_{\mathcal{C}})/k_B] \tag{3.23}$$

Remark. Now, $\mathcal{C}$ is specified both by its volume V and energy E. As before, one can expand $\log \Omega_B$ both in E_B and in V_B (around E_T and V_T) and take the limit $N_B \to \infty$.

$$P_{\mathcal{C}} \propto \exp\left[\frac{S_B(E_T,V_T)}{k_B} - \frac{E_{\mathcal{C}}}{k_B}\left.\frac{\partial S_B}{\partial E_B}\right|_{E_T,V_T} - \frac{V_{\mathcal{C}}}{k_B}\left.\frac{\partial S_B}{\partial V_B}\right|_{V_T,E_T} + \left(term \simeq \frac{1}{N_B}\right)\right] \tag{3.24}$$

Recalling that

$$\left.\frac{\mathrm{d}S}{\mathrm{d}V}\right|_E = \frac{P}{T} \qquad \text{with} \begin{cases} P_B \to P \\ T_B \to T \end{cases} \tag{3.25}$$

$$\Rightarrow P_{\mathcal{C}} \propto \exp\left[-\frac{E_{\mathcal{C}}}{k_B T} - \frac{PV_{\mathcal{C}}}{k_B T}\right] \tag{3.26}$$

If we normalize:

$$P_{\mathcal{C}} = \frac{e^{-\beta(E_{\mathcal{C}} + PV_{\mathcal{C}})}}{\Delta(T,P,N)} \tag{3.27}$$

where

$$\Delta(T,P,N) = \sum_{\mathcal{C}} e^{-\beta(E(\mathcal{C})+PV(\mathcal{C}))} \tag{3.28}$$

is called the **Gibbs partition function**.

Remark. Note that

$$\begin{aligned}\Delta(T,P,N) &= \sum_V e^{-\beta PV}\Big(\sum_{\substack{\mathcal{C}\\ V,N\text{fixed}}} e^{-\beta E_{\mathcal{C}}}\Big) = \sum_V e^{-\beta PV} Q(T,V,N)\\ &= \underbrace{\sum_V\sum_E}_{\substack{\text{fluctuating}\\ \text{variables}}} e^{-\beta(E+PV)}\Omega(E,V,N)\end{aligned}$$

By summing over all the microstates compatible with E and V:

$$P(E,V) = \frac{\Omega(E,V,N)}{\Delta(T,P,N)} e^{-\beta(E+PV)} \tag{3.29}$$

Remark.

$$\Delta(T,P,N) = \underbrace{\sum_E\sum_V e^{-\beta E-\beta PV}}_{\text{Laplace transform}}\Omega(E,V,N) = \sum_{E,V} e^{-\beta E-\beta PV+S(E,V,N)/k_B} \tag{3.30}$$

Classical systems (fluids)

$$\Delta(T,P,N) = \int_0^\infty \mathrm{d}V\, e^{-\beta PV}\left[\frac{1}{h^{3N}N!}\int \mathrm{d}\vec{\mathbf{p_1}}\ldots\mathrm{d}\vec{\mathbf{p_N}}\, e^{-\beta\mathcal{H}(p^N,r^N)}\right] \tag{3.31}$$

which implies

$$\Delta(T,P,N) = \int_0^\infty \mathrm{d}V\, e^{-\beta PV} Q(T,V,N) \tag{3.32}$$

that is the Laplace transform of the canonical partition function Q.

We also define

$$\beta P \equiv \frac{P}{T} = \frac{\partial S}{\partial V} \tag{3.33}$$

Remark. Let us remind that

Definition 3: Laplace transform

The Laplace transform of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is a unilateral transform defined by

$$F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$$

where s is a complex number frequency parameter with real numbers σ and ω:

$$s = \sigma + i\omega$$

Magnetic system

Ensemble in which both E and M can fluctuate. In particular, we have $\Omega(E,M)$ (with T_B and H_B fixed).

$$P_{\mathcal{C}} \propto e^{\frac{S_B(E_T,M_T)}{k_B} - \frac{E_{\mathcal{C}}}{k_B}\frac{\mathrm{d}S_B}{\mathrm{d}E_B} - \frac{M_{\mathcal{C}}}{k_B}\frac{\mathrm{d}S_B}{\mathrm{d}M_B}} \tag{3.34}$$

Since $\frac{\mathrm{d}S_B}{\mathrm{d}M_B} = -\frac{H_B}{T_B}$ and $\frac{\mathrm{d}S_B}{\mathrm{d}E_B} = \frac{1}{T_B}$:

$$\Rightarrow P_{\mathcal{C}} \propto \exp[-\beta(E_{\mathcal{C}} - HM_{\mathcal{C}})], \qquad T_B \to T, H_B \to H \tag{3.35}$$

The normalization function is:

$$\Delta(T,H,N) = \sum_{\mathfrak{e}} e^{-\beta(E_{\mathfrak{e}} - HM_{\mathfrak{e}})} = \sum_{E,M} e^{-\beta E + \beta MH}\Omega(E,M) \tag{3.36}$$

that is the *Gibbs partition function* for magnetic systems.

3.3.1 Saddle point approximation

The sum (3.30) can be approximated by the maximum of the integrand (this is fair for highly peaked functions):

$$\sum_{E,V} \exp[-\beta E - \beta PV + S(E,V,N)/k_B] \approx e^{-\beta E^* - \beta PV^* + S(E^*,V^*,N)/k_B}$$

where

$$\left(\frac{\mathrm{d}S(E^*,V^*,N)}{\mathrm{d}E}\right)_{V,N} = \frac{1}{T}, \qquad \left(\frac{\mathrm{d}S(E^*,V^*,N)}{\mathrm{d}V}\right)_{E,N} = \frac{P}{T}$$

this implies

$$-k_B T \ln \Delta(T,P,N) \simeq E^* + PV^* - TS$$

Hence, we define the **Gibbs free energy** as:

$$G(T,P,N) = -k_B T \ln \Delta(T,P,N) \tag{3.37}$$

3.4 Gran canonical ensemble

In this case N varies instead than V. Thus we have

$$\begin{aligned} P_{\mathfrak{e}} &= \exp\left[\frac{S_B(E_T - E_{\mathfrak{e}}, N_T - N_{\mathfrak{e}})}{k_B}\right] \\ &\sim \exp\left[\frac{S_B(E_T,N_T)}{k_B} - \frac{E_{\mathfrak{e}}}{k_B}\frac{\mathrm{d}S_B}{\mathrm{d}E_B} - \frac{N_{\mathfrak{e}}}{k_B}\frac{\mathrm{d}S_B}{\mathrm{d}N_B} + \left(\text{terms of order} \leq \frac{1}{V_B}\right)\right] \\ &= \frac{\exp[-\beta E_{\mathfrak{e}} + \beta\mu N_{\mathfrak{e}}]}{\Theta(T,V,\mu)} \end{aligned} \tag{3.38}$$

where

$$\Theta(T,V,\mu) = \sum_N \sum_{\substack{\mathfrak{e} \\ V,N\text{fixed}}} e^{-\beta(E_{\mathfrak{e}} - \mu N)} \tag{3.39}$$

is the **grancanonical partition function**.

Remark. Remember that

$$\frac{\mathrm{d}S}{\mathrm{d}E} = \frac{1}{T}, \qquad \frac{\mathrm{d}S}{\mathrm{d}N} = \frac{\mu}{T} \tag{3.40}$$

The *fugacity* is defined as:

$$z \equiv e^{\beta\mu} \tag{3.41}$$

and we rewrite

$$\Theta(T,V,\mu) = \sum_{N=0}^{\infty} z^N \Big(\sum_{\substack{\mathfrak{e} \\ V,N\text{fixed}}} e^{-\beta E_{\mathfrak{e}}} \Big) \tag{3.42}$$

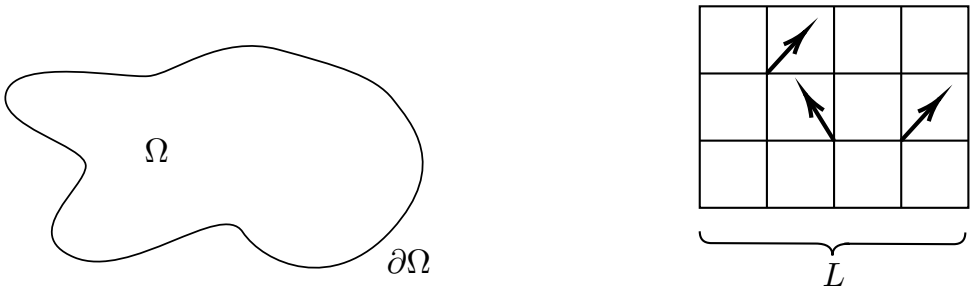
In principle, if one is able to compute the partition function is able to compute the thermodynamic quantitites.

Chapter 4

Statistical mechanics and phase transitions

4.1 Statistical mechanics of phase transitions

From the microscopic degrees of freedom, one compute the partition function in the appropriate ensemble, then the corresponding thermodynamic potential and from it all the thermodynamic properties of the system as *equilibrium phases* and, if present, *phase transitions*. Actually, until the '30 there were strong concerns about the possibility that statistical mechanics could describe phase transitions.



(a) Region Ω with boundary $\partial\Omega$.

(b) Magnetic system with characteristic length L.

Figure 4.1

Let us consider a system withing a region Ω of volume $V(\Omega)$ and boundary $\partial\Omega$ of area $S(\Omega)$. Denoting by L a characteristic lenght of the system

$$V(\Omega) \propto L^d, \quad S(\Omega) \propto L^{d-1}$$

where d is the spatial dimension where the system is embedded.

Remark. Space Ω can be either *discrete* or *continuous*.

Suppose that the system is *finite*. Formally, we can write

$$\mathcal{H}_\Omega = -\sum_n k_n \Theta_n \tag{4.1}$$

where

- k_n: are the coupling constants. In general, but not always, they are proportional to the *intensive thermodynamic variables*.
- Θ_n: is a linear, or higher order, combination of the dynamical microscopic degrees of freedom (local operators in quantum statistical mechanics).
- $k_n\Theta_n$: must obey the symmetry of the system. It is important that in principle the term satisfies the symmetry of the system. This is a master rule!

Remark. The symmetry of the system is related to the space of degrees of freedom, which in general is different from the embedding space Ω.

To fix the idea, let us consider two classical examples: the magnetic system and the fluid system.

4.1.1 Magnetic system (canonical)

The degrees of freedom are the *spins* lying on a Bravais lattice $\vec{\mathbf{S_i}}$, with $1 \leq i \leq N(\Omega)$, where the $N(\Omega)$ are the number of lattice sites. A configuration is the orientation of the spin in each site $\mathcal{C} = \{\vec{\mathbf{S_1}}, \dots, \vec{\mathbf{S_N}}\}$. We have:

$$\Theta_1 = \sum_i \vec{\mathbf{S_i}} \tag{4.2a}$$

$$\Theta_2 = \sum_{ij} \vec{\mathbf{S_i}} \cdot \vec{\mathbf{S_j}} \tag{4.2b}$$

We consider the trace operation, that is the sum over all possible values that each degree of freedom can assume:

$$\mathrm{Tr} \equiv \sum_{\{\mathcal{C}\}} \equiv \sum_{\vec{\mathbf{S_1}}} \sum_{\vec{\mathbf{S_2}}} \cdots \sum_{\vec{\mathbf{S_N}}} \tag{4.3}$$

where $\sum$ can also indicate an integration if values are continuous.

The *canonic partition function* is

$$Q_\Omega(T, \{k_n\}) = \mathrm{Tr}\big(e^{-\beta \mathcal{H}_\Omega}\big) \tag{4.4}$$

with $\beta \equiv \frac{1}{k_B T}$.

4.1.2 Fluid system (gran canonical)

Consider N particles in a volume V, with number density $\rho = N/V$. The $2dN$ degrees of freedom are

$$\{\mathcal{C}\} = \{(\vec{\mathbf{x_i}}, \vec{\mathbf{p_i}})_{i=1,\dots,N}\}$$

and

$$\Theta_1 = \sum_i \left[\frac{\vec{\mathbf{p_i}}^2}{2m_i} + U_1(\vec{\mathbf{x_i}}) \right] \tag{4.5a}$$

$$\Theta_2 = \frac{1}{2} \sum_{i>j} U(|\vec{\mathbf{x_i}} - \vec{\mathbf{x_j}}|) \tag{4.5b}$$

The trace operation is

$$\mathrm{Tr} \equiv \sum_{\{\mathcal{C}\}} = \sum_{N=0}^{\infty} \frac{1}{N!} \int \prod_{i=1}^{N} \frac{(\mathrm{d}\vec{\mathbf{p_i}})(\mathrm{d}\vec{\mathbf{x_i}})}{h^{dN}} \tag{4.6}$$

The *gran canonical partition function* is:

$$\mathcal{F}_\Omega = \mathrm{Tr}\Big(e^{-\beta(\mathcal{H}_\Omega - \mu N)}\Big) \tag{4.7}$$

For a generic partition function $Q_\Omega(T, \{k_n\})$, we can define the finite system *free energy* as

$$F_\Omega[T, \{k_n\}] = -k_B T \ln Q_\Omega(T, \{k_n\}) \tag{4.8}$$

The relation with thermodynamic is trough the *theromdynamic limit.*

Since the free energy is an extensive function,

$$F_\Omega \propto V(\Omega) \sim L^d$$

In general, one can write

$$F_\Omega[T, \{k_n\}] = V(\Omega) f_b[T, \{k_n\}] + S(\Omega) f_s[T, \{k_n\}] + O(L^{d-2}) \tag{4.9}$$

where $f_b[T, \{k_n\}]$ is the bulk free energy density.

Definition 4: Bulk free energy density

We define the *bulk free energy density* as

$$f_b[T, \{k_n\}] \equiv \lim_{V(\Omega)\to\infty} \frac{F_\Omega[T, \{k_n\}]}{V(\Omega)} \tag{4.10}$$

if the limit exists (to prove for each system) and does not depend on Ω.

For a system defined on a lattice we have

$$L(\Omega) \propto N(\Omega)^{1/d}, \quad V(\Omega) \propto N(\Omega)$$

$$f_b[T, \{k_n\}] = \lim_{N(\Omega)\to\infty} \frac{1}{N(\Omega)} F_N[T, \{k_n\}]$$

To get information on surface property of the system, let us calculate

$$f_s[T, \{k_n\}] \equiv \lim_{S(\Omega)\to\infty} \frac{F_\Omega[T, \{k_n\}] - V(\Omega) f_b[T, \{k_n\}]}{S(\Omega)} \tag{4.11}$$

4.1.3 Partition Function as Generating Function

Given the canonical partition function:

$$Z[T, \{k_m\}] = \mathrm{Tr}_{\{C\}}\, e^{-\beta E(C)} \tag{4.12}$$

we can compute the thermal averages at equilibrium as derivatives of Z. For instance the average of the energy $\langle \frac{1}{Z}\,\mathrm{Tr}\rangle[E(C)\exp(-\beta E(C))]$

4.1.4 Thermodynamic limit with additional constraints

For a fluid we cannot simply take the limit $V(\Omega) \to \infty$ by keeping N fixed, otherwise we will always get a infiinite system with *zero density*. One has to take also the limit $N(\Omega) \to \infty$ such that:

$$\frac{N(\Omega)}{V(\Omega)} \equiv \rho = const$$

In general, is not so easy to prove the existence of the limit and it depends on the range of the particle-particle interactions.

4.1.5 Statistical mechanics and phase transitions

Since all the thermodynamic information of a system can be obtained by the partition function, in principle, also the ones concerning the existence and nature of the phase transition must be contained in Z (or Q). On the other hand, we know from thermodynamic that phase transitions are characterized by singularities in the derivation of F. Also Z must display these singularities.

On the other hand, Z is a sum of exponentials

$$Z_\Omega = \mathrm{Tr}\big(e^{-\beta\mathcal{H}_\Omega}\big) \tag{4.13}$$

the exponentials are analytic functions everywhere (they converge), hence Z_Ω is analytic for Ω finite!

The question is: where do singularities come from? It is only in the thermodynamic limit that singularities in F and hence points describing phase transitions can arise!

For summarizing, there is no way out of this for producing singularities. The singularities will develop in the thermodynamic limits. For reaching singularities, we have to reach so precision in thermodynamic that we are not able to go exactly into the critical point. How can we relate singularities, geometrically, in the behaviour of the system?

4.2 Critical point and correlations of fluctuations

From thermodynamics, we know that, at the critical point, some response functions may diverge. Now, we show that this is a consequence of the onset of microscopic fluctuactions that are spatially correlated over long distances. To see this, let us compute the response of a ferromagnetic in presence of an external magnetic field H. The Gibbs partition function of a generic magnetic system is as equation:

$$Z_{\text{Gibbs}}[T, \{k_n\}] = \text{Tr}\Big(e^{-\beta(\mathcal{H}(\mathcal{C})-HM(\mathcal{C}))}\Big) = \sum_{M,E} e^{-\beta E+\beta HM}\Omega(E, M)$$

Remark. The term $(-HM)$ is the work done by the system against the external field H to mantain a given magnetization M.

$$\langle M\rangle = \left.\frac{\partial \ln Z_G}{\partial(\beta H)}\right|_T = \frac{1}{Z_G}\,\text{Tr}\Big[M(\mathcal{C})e^{-\beta(\mathcal{H}(\mathcal{C})-HM(\mathcal{C}))}\Big] \tag{4.14}$$

$$\chi_T = \frac{\partial\langle M\rangle}{\partial H} = \left\{\frac{\beta}{Z_G}\,\text{Tr}\left[M^2(\mathcal{C})e^{-\beta\mathcal{H}+\beta HM}\right] - \frac{\beta}{Z_G^2}\left[\text{Tr}\left[M(\mathcal{C})e^{-\beta\mathcal{H}+\beta HM}\right]\right]^2\right\} \tag{4.15}$$

Hence,

$$\chi_T = \frac{1}{k_B T}\Big(\langle M^2\rangle - \langle M\rangle^2\Big) \tag{4.16}$$

The thermodynamic response function χ_T in statistical mechanics is related to the variance of the magnetization.

We can relate the above expression with the correlation of the microscopic by performing a coarse-graining of the system, where the magnetization $M(\mathcal{C})$ can be computed as an integral

$$M(\mathcal{C}) = \int \mathrm{d}^3\vec{\mathbf{r}}\, m(\vec{\mathbf{r}}) \tag{4.17}$$

where $m(\vec{\mathbf{r}}$ is defined through a coarse graining approach as the local average on a block of an infinitesimal subsystem. Hence,

$$k_B T\chi_T = \int \mathrm{d}\vec{\mathbf{r}}\,\mathrm{d}\vec{\mathbf{r}'}\left[\Big\langle m(\vec{\mathbf{r}})m(\vec{\mathbf{r}'})\Big\rangle - \langle m(\vec{\mathbf{r}})\rangle\Big\langle m(\vec{\mathbf{r}'})\Big\rangle\right] \tag{4.18}$$

Let us assume the *translational symmetry*:

$$\begin{cases}\langle m(\vec{\mathbf{r}})\rangle = m & \text{homogeneous}\\ \Big\langle m(\vec{\mathbf{r}})m(\vec{\mathbf{r}'})\Big\rangle \equiv G(\vec{\mathbf{r}}-\vec{\mathbf{r}'}) & \text{two-point correlation function}\end{cases} \tag{4.19}$$

Instead, let us consider the *connected correlation function*, i.e. the correlation function of the fluctuations $\delta m = m - \langle m\rangle$:

$$\Big\langle m(\vec{\mathbf{r}})m(\vec{\mathbf{r}'})\Big\rangle_c \equiv \Big\langle (m(\vec{\mathbf{r}}) - \langle m(\vec{\mathbf{r}})\rangle)\Big(m(\vec{\mathbf{r}'}) - \Big\langle m(\vec{\mathbf{r}'})\Big\rangle\Big)\Big\rangle = G(\vec{\mathbf{r}}-\vec{\mathbf{r}'}) - m^2 \tag{4.20}$$

Given the translational invariance, one can centre the system such that its centre of mass coincides with the origin

$$\vec{\mathbf{r}}_{CM} \Rightarrow \vec{\mathbf{r}}_0 \equiv \vec{\mathbf{0}}$$
$$\Rightarrow \int \mathrm{d}\vec{\mathbf{r}}\int \mathrm{d}\vec{\mathbf{r}'}\,[G(\vec{\mathbf{r}}-\vec{\mathbf{r}}_0) - m^2]$$

The integration over $\vec{\mathbf{r}'}$ gives the volume $V(\Omega)$ of the system:

$$\underbrace{k_B T\chi_T}_{\substack{\text{response}\\ \text{function}}} = V(\Omega)\int \mathrm{d}\vec{\mathbf{r}}\quad \underbrace{\langle m(\vec{\mathbf{r}})m(\vec{\mathbf{r}}_0)\rangle_c}_{\substack{\text{correlation function}\\ \text{of the fluctuations}\\ \text{of the local magnetization}}} \tag{4.21}$$

The equation is called the **Fluctuation-Response Relation**.

How $G_c(\vec{\mathbf{r}})$ behaves? In general, one has

$$G_c(\vec{\mathbf{r}}) \sim e^{-|\vec{\mathbf{r}}|/\xi} \tag{4.22}$$

meaning that for $|\vec{\mathbf{r}}| > \xi$ the fluctuations are uncorrelated, where ξ is the *correlation length*. The correlation length is related to the correlation function. In general, it is finite but, if you approach T_c, it diverges. In fact, at the critical point this correlation will expand in the whole space and reaches the size of all the system, in other words, it goes to infinity ($\xi \to \infty$). When ξ will diverge, there will not be anymore the exponential and the integral cannot be keeped finite.

We can estimate the integral:

$$\int G_c(r)\, dr = 4\pi \int |r|^2 G_c(|r|)\, d|r| \tag{4.23}$$

Since $G_c(|r|) \propto \exp\{-|r|/\xi\}$

$$4\pi \int |r|^2 e^{-|r|/\xi}\, d|r| \tag{4.24}$$

On the other hand

$$\int_0^\infty x^2 e^{-\frac{x}{\xi}}\, dx = 2\xi^3 \tag{4.25}$$

So if we let g be the value of G_c for $|\vec{\mathbf{r}}| < \xi$:

$$k_B T \chi_T \leq V g \xi^3$$

where there is an inequality because we are understimating the integral..

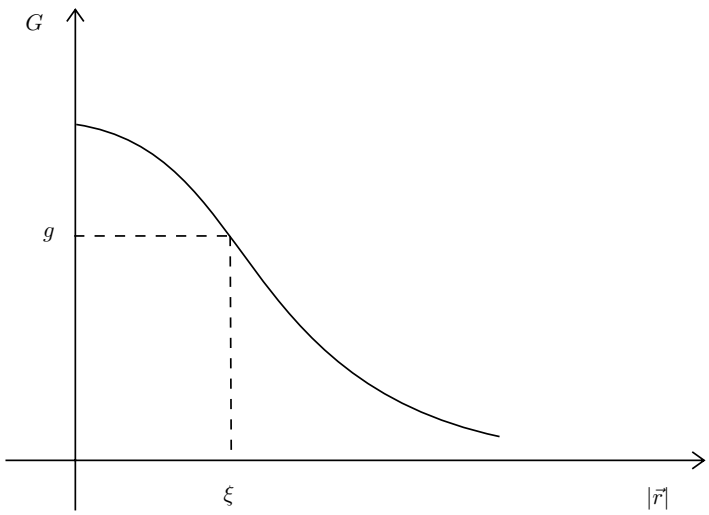


Figure 4.2: Plot of the two-point correlation function, G.

Rearranging the terms, we obtain

$$\frac{k_B T \chi_T}{V} \leq g\xi^3 \tag{4.26}$$

Hence, if χ_T diverges at the critical point it implies $\xi \to \infty$.

In particular, one can see that for $H = 0$ and $T \to T_c^{\pm}$:

$$\xi_{\pm}(T, H = 0) \sim |t|^{-\nu_{\pm}} \tag{4.27}$$

where $\nu_+ = \nu_- = \nu$ is the **Correlation Length Critical Exponent**.

Remark. It does not derive from thermodynamic considerations, as it is due to the collective behaviour of the units.

Scaling is often used as the most general definition of a critical point.

One can also show that at $T = T_c$ (i.e. $t = 0$)

$$G_c(r) \sim \frac{1}{r^{d-2+\eta}} \tag{4.28}$$

where η is the correlation critical exponent.

Remark. The formula is a power law decay instead than exponential.

4.2.1 Critical opalescence: experimental evidence of correlation length divergence

Microscopic correlations can be proved by *scattering experiments*. For instance, near the critical liquid-gas mixture, the fluid appears "milky", while away from T_c it is transparent. This means that the system is diffusing light, which has $\lambda = 500nm$ though, which is much larger than typical atomic distances ($a \approx 1nm$). The scattering of light is caused by density fluctuations, that need to be correlated over long distance to make ξ comparable to λ.

4.3 Finite size effects and phase transitions

Actually, the thermodynamic limit is a mathematical trick and in real systems it is never reached. Is it then physically relevant?

If we had instruments with *infinite* precision each change of the physical properties of a system would occur within a finite range, therefore we would observe a smooth crossover instead than a singularity. In this respect, the notion of correlation length ξ is extremely important.

To illustrate this point, let us consider the gas-liquid system in proximity of its critical point ($T \sim T_c$). If we approach T_c from the gas phase, there will be fluctuations of ρ with respect to ρ_G, $\Delta\rho = \rho - \rho_G$, due to the presence of denser droplets (liquid) in the continuum gas phase. These droplets will have different diameters, but the average size would be ξ, where it is the typical size of the liquid droplets. Clearly $\xi = \xi[T]$ and, in proximity of the critical point $\xi \overset{t\to 0}{\sim} |t|^{-\nu}$.

On the other hand, in a finite system, ξ cannot diverge since is bounded above, $\xi \leq L$, where L is the linear system size.

As $T \to T_c$, where ξ should be larger than the system size, the behaviour of the system should deviate from the one expected by the theory that is obtained in the limit $L \to \infty$. How far the real system would be from the critical point $t = 0$ where singularities develop? Let us try to give an estimate of this deviation.

Let us consider a system of size $L = 1\,\text{cm}$ and

$$t \equiv (T - T_c)/T_c, \qquad \xi \sim \xi_0 t^{-\nu}$$

Let us assume that the lattice distance is $\xi_0 = 10\,\text{Å}$. Hence,

$$t \sim \left(\frac{\xi}{\xi_0}\right)^{-1/\nu} \sim \left(\frac{L}{10\,\text{Å}}\right)^{-1/\nu} \sim (10^7)^{-1/\nu} \tag{4.29}$$

In the next chapters, we will see that $\nu < 1$ and close to $1/2$, hence:

$$t \sim (10^7)^{-2} = 10^{-14}$$

Therefore we have $t \approx 10^{-14}$ as a distance from T_c.

This estimate suggests that the experimental instrument that measures temperature must have a precision of 10^{-14} to see deviations from the results obtained in the thermodynamic limit.

4.4 Numerical simulations and phase transitions

In this case, the size L of the simulated system is few multiples of ξ_0 and the finite-size effects of the simulated data can strongly affect the location and the scaling laws of the phase transition under numerical investigation. Finite size scaling analysis of the numerical data is needed.

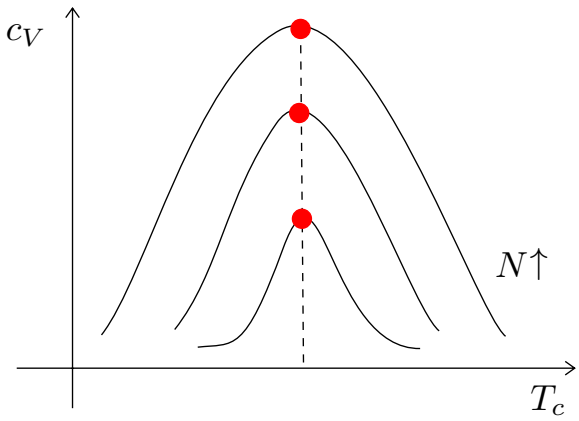


Figure 4.3: (T_c, C_V) plot at different N.

We can find the critical point by doing Montecarlo simulation. Supposing a Montecarlo simulation of a Ising model, for which there is no an analytic solution and compute the energy. Try to extrapolate for example the position of the peak as N increases. If we start to see the behaviour, something is happening. There are two approaches we can use.

The first approach is studying the system by looking for all the details. An example could be a protein, that interact with other proteins; in this case we can look at all the electrons (or atoms). Nevertheless, even if we thought at the simple protein that exists, there would be a lot of degrees of freedom.

For doing a simulation, if we are interested in long time behaviour and in large scale behaviour, details are not important. What it is important are symmetries, ranges of interaction. Therefore, we can forget about all the details. We can introduce the effective potentials as Van der Waals or Lenard Jones potential and studying collective effects. This is the second approach.

Chapter 5

Role of the models in statistical mechanics

5.1 Role of the models

Which is the role of models in statistical mechanics? There are two possible approaches:

1. The model must describe the real system in a very detailed way. The maximum number of details and parameters to be tuned are included. The *pro* is the closer to the real specific system (faithfull description). The *drawback* is that the model is so complicated that no analytical solution is possible. Moreover, even numerically, these models can be studied for very short times and small sizes. An example is the simulation of the folding dynamics that can be performed for few nanoseconds. On the other hand, the introduction of many details are often not crucial if one is interested in large scale properties.

2. Try to introduce (coarse-graining approach) the most simple model that satisfies few essential properties of the real system such as its symmetries, dimensionality, range of interactions etc. Since most of the microscopic details are integrated, these models cannot describe the full physics of a specific system but they can reproduce its main features. Moreover, these models can be studied numerically and, to some extent, also analytically (exact solution).

It is the latter approach that we shall take here. Let us start by introducing what is, perhaps, the most paradigmatic model in the statistical mechanics of phase transition, the *Ising model*.

5.2 The Ising model

Suggested by Lenz to Ising for his PHD thesis (1925), it is supposed to describe a magnetic system that undergoes a transition between a paramagnetic and a ferromagnetic phase. In $d = 1$ the model was solved exactly by Ising. Unfortunately, he found that for $T > 0$ the model does not display a phase transition.

The wrong conclusion was that this model was not able to describe a phase transition. In fact, it turns out that, for $d > 1$, the model does display a paramagnetic-ferromagnetic phase transition.

Let us first discuss some general feature of the model for any dimension d.

5.2.1 d-dimensional Ising model

For hypercubic lattice with given $N(\Omega)$ sites $\{i\}_{i=1,\dots,N(\Omega)}$ and linear size $L(\Omega)$, we have

$$N(\Omega) = L^d$$

The microscopic degrees of freedom are the spins S_i, defined at each i-esim lattice site. Each spin can assume the values $S_i = \pm 1$, that means that at each site the possible values are the spin up or down. For a lattice with $N(\Omega)$ spins, there are $2^{N(\Omega)}$ possible configurations.

Remark. Since we do not consider the spin as a vector, this is a model for a strongly anysotropic ferromagnet (along a given direction).

The minimal model that can try to capture the interaction between the spin is the following. Suppose to have also an external magnetic field H_i (it values depends on the site i). One can consider interactions between spins whose strength are described by functions $J_{ij}, k_{ijk}, \ldots$. For instance, there is a coupling that derives from electrons coupling

$$J_{ij} = f(|\vec{\mathbf{r_i}} - \vec{\mathbf{r_j}}|)$$

The physical origin is the overlap between the electronic orbitals of the neighbouring atoms forming the Bravais lattice. Remember that a term as $\sum_i S_i$ is not correlated, while we need an interaction for describing the model.

A general Hamiltonian of the model can be written as:

$$\mathcal{H}_\Omega(\{S_i\}) = \sum_{ij} J_{ij} S_i S_j - \sum_i H_i S_i - \sum_{ijk} S_i S_j S_k + \ldots$$

Standard Ising model one keeps only the two-body interactions:

$$\mathcal{H}_\Omega(\{S_i\}) = -\frac{1}{2} \sum_{i,j}^{N} J_{ij} S_i S_j - \sum_{i=1}^{N} H_i S_i \tag{5.1}$$

where the first term represents a two body interaction that is a quadratic term, while the second term is a one body interaction. We have put the minus because we want to minimize the energy, but it depends on the sign of J.

For this model, the sum over all configurations on trace is given by

$$\mathrm{Tr} \equiv \sum_{S_1=\pm 1} \sum_{S_2=\pm 1} \cdots \sum_{S_N=\pm 1} \equiv \sum_{\{S\}}$$

Our problem is to find the partition function with N sites, which depends on T and in principle depends on the configuration given (it is fixed both for H and J!). Hence, the *canonical partition function* is given by

$$Z_\Omega(T, \{H_i\}, \{J_{ij}\}) = \mathrm{Tr}\, e^{-\beta \mathcal{H}_\Omega(\{S\})} \tag{5.2}$$

and the corresponding *free-energy*,

$$F_\Omega(T, \{H_i\}, \{J_{ij}\}) = -k_B T \ln Z_\Omega \tag{5.3}$$

The *bulk limiting free energy* is:

$$f_b(T, \{H_i\}, \{J_{ij}\}) = \lim_{N\to\infty} \frac{1}{N} F_\Omega \tag{5.4}$$

How do we know that the above limit does exist? It must be proven. The surface is not important in the bulk limit. Note that we are assuming that the *interaction between the spin is a short range force*, it is not as the size of the system.

For this model it is possible to show that the limit exists if

$$\sum_{j\neq i} |J_{ij}| < \infty \tag{5.5}$$

Remark. In general what determines the existence of the limit of these spin models are the *dimension* d and the *range* of the spins *interactions*.

For example it is possible to show that, if

$$J_{ij} = A|\vec{\mathbf{r_i}} - \vec{\mathbf{r_j}}|^{-\sigma} \tag{5.6}$$

so it is a long range interaction, the limit exists when

$$\sigma > d$$

Remark. If the interaction is *dipolar*, since it decades as $1/r^3$, for the case $d = 3$ the limit does not exists. However, it is still possible to prove the existence of the limit for this case if one assumes that not all dipoles are fully aligned.

Assuming that the thermodynamic limit exists, we now look at some additional rigorous results on the limiting free energy and its derivatives.

5.2.2 Thermodynamic Limit

For simplicity, let us consider the case in which the external magnetic field is homogeneous, i.e. $H_i \equiv H$, and the spin-spin interaction is only between spins that are nearest-neighbours (n.n.) on the lattice:

$$J_{ij} = \begin{cases} J & \text{if } i \text{ and } j \text{ are n.n.} \\ 0 & \text{otherwise} \end{cases} \tag{5.7}$$

Now, the model is very simple:

$$-\mathcal{H}_\Omega(\{S\}) = J \sum_{\langle ij \rangle}^{N(\Omega)} S_i S_j + H \sum_i^{N(\Omega)} S_i \tag{5.8}$$

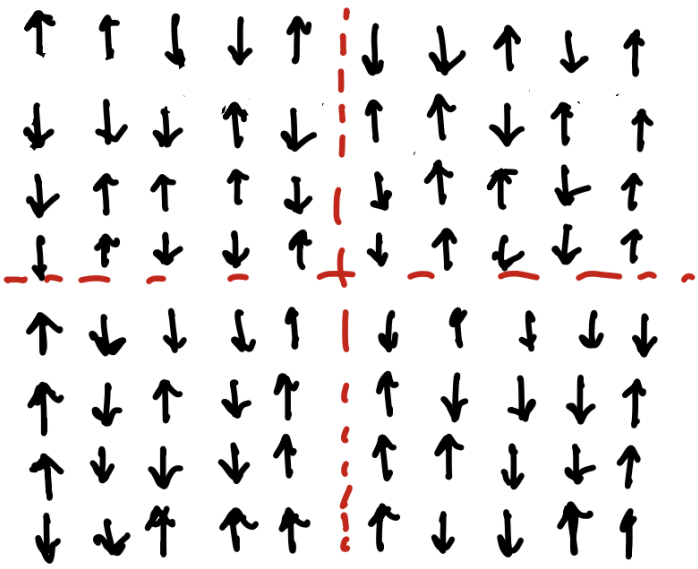
where the notation $\langle ij \rangle$ means a double sum over i and j, with the constraint that i and j are nearest-neighbours.

To prove the existence of the thermodynamic limit of the Ising Model let us consider the case in which $H_i = H = 0$ for all nodes. So our Hamiltonian becomes:

$$\mathcal{H}_N(\{s\}) = -J \sum_{\langle ij \rangle} s_i s_j$$

To simplify the drawing let us prove it for $d = 2$.

Proof. Given a square lattice with $N = 4n^2$ spins let us partition it into 4 identical subsystems each with n^2 spins:



If we neglect the surface interaction between the 4 subsystems we make an error in the estimate of the total energy ranging between $-4Jn$ (boundary spins are all parallel) and $4Jn$ (boundary spins are all anti-parallel). Since the number of configurations is not modified by this partition if we assume that the subsystems are independent:

$$(Z_{n^2})^4 e^{-4\beta Jn} \leq Z_{(2n)^2} \leq (Z_{n^2})^4 e^{4\beta Jn}$$

With a free-energy density of a single subsystem:

$$f_{n^2} = -k_B T \ln \left(\frac{Z_{n^2}}{n^2} \right)$$

And for all the system:

$$f_{(2n)^2} = -k_B T \ln \left(\frac{Z_{(2n)^2}}{4n^2} \right)$$

By taking $k_B T \ln$ of the inequality above and dividing by the total "volume" $N = 4n^2$ we have

$$\frac{4}{\beta} \ln\left(\frac{Z_{n^2}}{4n^2}\right) - \frac{4\beta J_n}{\beta 4n^2} \leq f_{(2n)^2} \leq \frac{4\ln(Z_{n^2})}{\beta 4n^2} + \frac{4\beta J_n}{\beta 4n^2}$$

Hence

$$f_{n^2} - \frac{J}{n} \leq f_{(2n)^2} \leq f_{n^2} + \frac{J}{n}$$

That can be written has:

$$\mid f_{(2n)^2} - f_{n^2} \mid \leq \frac{\text{const}}{n} = \text{const} \cdot \frac{n}{n^2} \sim \frac{L}{S}$$

With $L = n$ as the "perimeter" and $S = n^2$ as the area. The difference is due to boundary effects. If f_{n^2} satisfies the last inequality it is a Cauchy Sequence so it converges to a finite limit.

The argument is easy to generalise to any dimensions: if $d = 3$ consider a cube of $8N^3$ spins and partition it into 8 sub-cubes, each of volume N^3 so $\frac{L}{S}$ becomes $\frac{S}{V}$. ■

Remark. If $\mathcal{H}(\{s\}) = -\frac{J}{2}\sum_{i,j} s_i s_j$ where the sum is over all pairs independently of their distance at $T = 0$ all the spins are aligned and $E \sim n^4$ so our limit becomes $\lim_{n\to\infty} \frac{E}{n^2} \sim n^2 \to \infty$

5.2.3 Order Parameter of the Ising Model

Let us start again from the Hamiltonian:

$$-\mathcal{H}(\{S\}) = J \sum_{\langle ij \rangle}^{N} S_i S_j + H \sum_{i}^{N} S_i$$

One can formally express the Average Total Magnetization as

$$M = \left\langle \sum_{i=1}^{N} S_i \right\rangle$$

In the case in which H is uniform we have

$$M(\beta) = \left\langle \sum_{i=1}^{N} s_i \right\rangle = \frac{1}{Z_N} \operatorname{Tr}_{\{s\}} \left(\sum_i s_i\right) e^{-\beta \mathcal{H}_N} = \frac{1}{\beta} \frac{\partial \ln(Z_N)}{\partial H}\bigg|_\beta$$

With Z_N the generating function. In general one is interested in the magnetisation density or per spins, so

$$m(\beta, H) = \frac{M(\beta, H)}{N} = \frac{1}{N\beta} \frac{\partial \ln(Z_N)}{\partial H}\bigg|_\beta = -\frac{1}{N} \frac{\partial F_N}{\partial H}\bigg|_\beta = -\frac{\partial f_N}{\partial H}\bigg|_\beta$$

With $m(\beta, H)$ being the **Order Parameter** of the Ising Model.

Note that for $H_i = H$, m does not depend on i (it's homogeneous) and, since $M = \sum_i s_i$ the partition function

$$Z_N(\beta, J, H) = \operatorname{Tr}_{\{s\}} \exp\left(\beta J \sum_{\langle ij \rangle} s_i s_j + \beta H M\right)$$

coincides with the Gibbs partition function (and f_n with the finite size Gibbs free-energy density g_N).

5.2.4 Ising model and $\mathbb{Z}^2$ symmetry.

The symmetry of the system in sense of the Hamiltonian is: we can invert the value of the S and the Hamiltonian does not change. It is valid when $H = 0$, otherwise it is not true. Let us see the $\mathbb{Z}^2$ symmetry and the following interesting relation:

Lemma 2

$\forall$ function Φ of the configuration $\{S_i\}$, the following relation holds:

$$\sum_{\{S_i=\pm 1\}} \Phi(\{S_i\}) = \sum_{\{S_i=\pm 1\}} \Phi(\{-S_i\}) \tag{5.9}$$

this is true for all function of the spin.

Now, we consider the Hamiltonian of the Ising model:

$$-\mathcal{H}_\Omega = J \sum_{\langle ij \rangle}^{N(\Omega)} S_i S_j + H \sum_i^{N(\Omega)} S_i$$

Clearly,

$$\mathcal{H}(H, J, \{S_i\}) = \mathcal{H}_\Omega(-H, J, \{-S_i\}) \tag{5.10}$$

This is a spontaneous broken symmetry.

Hence,

$$\begin{aligned} Z_\Omega(-H, J, T) &= \sum_{\{S_i=\pm 1\}} \exp[-\beta \mathcal{H}_\Omega(-H, J, \{S_i\})] \underset{(5.9)}{=} \sum_{\{S_i=\pm 1\}} \exp[-\beta \mathcal{H}_\Omega(-H, J, \{-S_i\})] \\ &\underset{(5.10)}{=} \sum_{\{S_i=\pm 1\}} \exp[-\beta \mathcal{H}_\Omega(H, J, \{S_i\})] = Z_\Omega(H, J, T) \end{aligned} \tag{5.11}$$

Taking $-k_B T \ln$, we got:

$$F_\Omega(T, J, H) = F_\Omega(T, J, -H) \tag{5.12}$$

If we take the thermodynamic limit $\lim_{N\to\infty} \frac{1}{N}$, we have

$$\Rightarrow f_b(T, J, H) = f_b(T, J, -H) \tag{5.13}$$

and it means that the *free energy density is an even function of H*!

Remark. From the finite-size relation (5.12), one can show that a *finite-size* Ising model does not display a transition to a ferromagnetic phase (for all dimension d). Indeed,

$$N(\Omega)M(H) = -\frac{\partial F(H)}{\partial H} \underset{(5.12)}{=} -\frac{\partial F(-H)}{\partial (H)} = \frac{\partial F(-H)}{\partial (-H)} = -N(\Omega)M(-H) \tag{5.14}$$

Therefore:

$$M(H) = -M(-H), \quad \forall H \tag{5.15}$$

If $H = 0$, we have $M(0) = -M(0)$, that is valid if and only if $M(0) = 0$!

The magnetization of a finite system is, at $H = 0$, always zero. This is simply consequence of the symmetry argument shown above. We have not a phase transition.

Hence, it is only in the thermodynamic limit, where the symmetry is spontaneously broken, that the model displays a transition.

For resuming, although the Hamiltonian is invariant with respect to the transformation $H \to -H, \{S_i\} \to \{-S_i\}$, the thermodynamic state is not. This situation is called *spontaneous symmetry breaking.*